CPE 334 Software Engineering

LAB 04 : TokTickIT Actions Taken, Dashboards, and Final Regression.

Answer Part 1: Git Use with Engineering Workflow

1.1 Git Workflow

Network graph

Timeline of the most recent commits to this repository and its network ordered by most recently pushed to and updated daily.

```
* main (Merge PR #45: Release Lab 04 Final Sprint Release) |\ | * lab4-staging (Release Candidate:
all 6 issues integrated & verified) |/| * | feature/lab4-issue-6-hardening (Merge PR #44:
regression, accessibility polish) | | * | feature/lab4-issue-5-dashboards (Merge PR #43: IT Staff &
Requester dashboards) | | * | feature/lab4-issue-4-ticket-workflow (Merge PR #42: lifecycle matrix,
resolution gate) | | * | feature/lab4-issue-3-actions-ui (Merge PR #41: Actions Taken UI & modal) |
| * | feature/lab4-issue-2-actions-foundation (Merge PR #40: schema migration, seed, APIs) | | * |
feature/lab4-issue-1-spec (Merge PR #39: Sprint contract specification & test plan) |/ * commit:
init lab4-staging from main
```

Figure 1.1: GitHub Network graph - feature branches merged into lab4-staging, then into main.

Note: GitHub's Network graph had not refreshed at the time of capture

1.2 GitHub Project/Kanban

Backlog	Specified	Done / PR	PR Review	Done (All Sprint 4) ✅
(empty)	(empty)	(empty)	(empty)	**Issue #33:** Sprint 4 engineering contract (PR #39)
				Issue #34: Actions Taken foundation (PR #40)
				Issue #35: Actions Taken UI (PR #41)
				Issue #36: Ticket workflow & gate (PR #42)
				Issue #37: Role Dashboards (PR #43)
				Issue #38: Final Hardening & DoD (PR #44)

Figure 1.2: Project board - all Sprint 4 Issues in Done.

1.3 reviewer.md

Link : https://github.com/aranedlek/toctickit/blob/main/docs/lab-04/reviewer.md

Lab 4 — Peer Review Record

Author: Aran Edlek — 67070505230 — GitHub: `aranedlek`

Peer Reviewer: Jinjutha Nannarumit — 67070505232 — GitHub: `Ponatinylilbug`

Pull Requests I Authored

PR	Issue	Branch	Work	Reviewer Verdict
#39	#33	`feature/lab4-issue-1-spec`	Sprint 4 engineering contract (specification, tests, UI and API specifications)	Approved ✅
#40	#34	`feature/lab4-issue-2-actions-foundation`	Actions Taken foundation: ActionTaken model, seed, REST APIs, test database	Approved ✅
#41	#35	`feature/lab4-issue-3-actions-ui`	Actions Taken UI on Ticket Details: list, accessible modal, role restrictions	Approved ✅
#42	#36	`feature/lab4-issue-4-ticket-workflow`	Ticket workflow: resolution gate, status history, optimistic concurrency control	Approved ✅
#43	#37	`feature/lab4-issue-5-dashboards`	Requester and IT Staff Dashboards: backend aggregations, drill-down links	Approved ✅
#44	#38	`feature/lab4-issue-6-hardening`	Regression & performance tests (97 tests), accessibility polish, documentation	Approved ✅

Review Dialog Summary:

- **PR #39 (Spec):** Reviewer: "*Specification ครบถ้วน แนะนำให้เตรียม seed data ตั๋วที่มี 0, 1 และหลาย actions taken*" → Author: "***รับทราบและเพิ่มใน seed เรียบร้อยครับ***"
- **PR #40 (Actions Foundation):** Reviewer: "***โมเดล** ActionTaken แยกจาก InternalNote ถูกต้อง เทสต์ผ่านทั้งหมด อนุมัติครับ*"
- **PR #41 (Actions UI):** Reviewer: "*Modal มี validation ตรงตาม BR-03, BR-04 ครบถ้วน อนุมัติครับ*"
- **PR #42 (Workflow):** Reviewer: "*State transition matrix และ OCC แบบเนอร์ทำงานถูกต้อง อนุมัติครับ*"
- **PR #43 (Dashboards):** Reviewer: "***แดชบอร์ดทั้งสอง** Role สวยงามและค่านวณจาก DB aggregation ถูกต้อง อนุมัติครับ*"
- **PR #44 (Hardening):** Reviewer: "*เทสต์ 97 ข้อผ่าน 100% และคีย์บอร์ด accessibility ครบถ้วน อนุมัติ merge เข้า staging และ main ครับ*"

Figure 1.3: Rendered docs/lab-04/reviewer.md - reviewer identity and PR links.

1.4 README and .gitignore

TokTickIT

TokTickIT is a full-stack IT Service Desk project for CPE334.

The repository contains work from Lab 1 to Lab 4. Lab 2 added the Requester-facing ticket workflow, Lab 3 added login, three roles (Requester, IT Staff, Administrator), the IT Staff ticket workflow, and user management. Lab 4 adds Actions Taken, the final ticket workflow (resolution gate, status history, state update check), and a Dashboard for each role.

Tech Stack

- **Frontend:** React + TypeScript + Vite + Vanilla CSS (Zen Green Design System)
- **Backend:** Node.js + Express + TypeScript
- **Database:** PostgreSQL + Prisma ORM
- **Testing:** Vitest + Supertest + React Testing Library + Playwright

Repository Structure

```
toktickit/
├── frontend/
│   ├── src/
│   │   ├── components/       # ActionsTakenSection, UI components
│   │   ├── pages/            # StaffDashboard, RequesterDashboard, Tickets
│   │   └── tests/lab-04/     # Component unit & integration tests
├── backend/
│   ├── prisma/               # schema.prisma, migrations, seed.ts
│   ├── src/                  # routes, middleware, controllers
│   └── tests/lab-04/         # Backend API tests
├── docs/lab-04/              # specification.md, tests.md, ui-spec.md, reviewer.md
├── e2e/lab-04/               # Playwright E2E test suites
└── scratch/                  # Utilities, reports, scripts
```

Setup

Clone the Repository:

```
git clone https://github.com/aranedlek/toctickit.git
cd toctickit
```

Backend Setup & Migration:

```
cd backend
npm install
npx prisma migrate dev
npx prisma db seed
npm run dev   # Server starts on http://localhost:3000
```

Figure 1.6: Rendered README.md (Part 1: Overview, Tech Stack, Structure, Setup)

Frontend Setup & Seeded Accounts (Local Development Only)

```
cd frontend
npm install
npm run dev   # Client starts on http://localhost:5173
```

Every seeded account uses the initial password `Password123!`.

Role	Name	Email	State
Requester	Aran Edlek	`aran@example.com`	Active (Ready)
Requester	Anya Suphan	`anya@example.com`	Active
Requester	Ben Rattana	`ben@example.com`	Active
Requester	Eve Inactive	`eve@example.com`	Inactive
IT Staff	IT Staff One	`it1@example.com`	Active (Ready)
IT Staff	IT Staff Two	`it2@example.com`	Active
IT Staff	IT Staff Inactive	`it-inactive@example.com`	Inactive
Administrator	Admin User	`admin@example.com`	Active (Ready)

Demo (Lab 4)

1. **IT Staff Dashboard:** Log in as IT Staff (`it1@example.com`). The dashboard shows operational metric cards (Unassigned, Open, In Progress, Waiting on Requester, My Assigned). Clicking any card navigates to the pre-filtered Ticket Queue.
2. **Actions Taken:** Open Ticket Detail. View the parent-child Actions Taken list. Click "+ Record Action Taken", fill action description and result. Toggle "Follow-Up Required" to test mandatory follow-up validation.
3. **Resolution Gate:** Open tickets cannot jump directly to Resolved without completing active operational tasks. Requesters submitting resolution remain strictly advisory.
4. **Optimistic Concurrency Control:** Two concurrent editors on the same ticket receive HTTP 409 Conflict with a reload prompt when versions mismatch.
5. **Requester Dashboard:** Log in as Requester (`aran@example.com`). View personalized request counts, recent activity, and zero-metric empty states.

Figure 1.6: Rendered README.md (Part 2: Seeded Accounts & Demo Workflows)

Testing and Regression Execution

TokTickIT maintains rigorous test-driven verification across all layers:

Backend API & Regression Tests (Vitest):

```
cd backend
npm test   # Executes 64 automated tests across 11 test files (100% pass)
```

Frontend Component & Accessibility Tests (Vitest):

```
cd frontend
npm test   # Executes 33 automated tests across 9 test files (100% pass)
```

End-to-End Test Suite (Playwright):

```
npx playwright test   # Full user journeys across desktop, tablet, and mobile
```

Environment Variables

Variable	Description / Default
`DATABASE_URL`	PostgreSQL database connection string
`PORT`	API port (Default: `3000`)
`CLIENT_ORIGIN`	Allowed frontend CORS origin (Default: `http://localhost:5173`)
`SESSION_SECRET`	JWT encryption key for authenticating HTTP cookies

Figure 1.6: Rendered README.md (Part 3: Testing Instructions & Environment Configuration)

```
# dependencies
node_modules/
# env & secrets
.env
*.env
.env.test
!.env.example
# build output
dist/
build/
# prisma
backend/prisma/*.db
# logs & OS
*.log
.DS_Store
# uploaded attachment files
backend/uploads/*
!backend/uploads/.gitkeep
# playwright output
test-results/
playwright-report/
```

Figure 1.7: .gitignore

1.5 Repository Structure

Lab 4 artifacts are organized in dedicated `lab-04` folders separated by purpose: specifications and review records in `docs/lab-04`, backend API/integration tests in `backend/tests/lab-04`, frontend UI tests in `frontend/src/tests/lab-04`, end-to-end tests in `e2e/lab-04`, and UI evidence in `docs/lab-04/screenshots`.

```
docs/lab-04/
├── ai-use.md
├── api-spec.md
├── reviewer.md
├── specification.md
├── tests.md
├── ui-spec.md
└── screenshots/
backend/src/routes/
├── actionsTaken.ts
├── dashboard.ts
└── ticketWorkflow.ts
frontend/src/
├── components/ActionsTakenSection.tsx
├── pages/staff/StaffDashboard.tsx
└── pages/requester/RequesterDashboard.tsx
e2e/lab-04/
├── actions-taken-flow.spec.ts
├── dashboards.spec.ts
└── ticket-resolution.spec.ts
```

Figure 1.8: Repository directory structure

Answer Part 2: Spec DD

2.1 rendered docs/lab-04/specification.md

Link : https://github.com/aranedlek/toctickit/blob/main/docs/lab-04/specification.md

2.1.1 Functional Requirements

4. Functional Requirements

Actions Taken

- **FR-01:** IT Staff and Administrators can create an Action Taken on a Ticket with Action Date/Time, Action Description, Result, Follow-Up Required, Follow-up Note, and Attachment Notes.
- **FR-02:** An action is assigned to an active IT Staff member or Administrator, who may be different from the Ticket Owner.
- **FR-03:** Staff Ticket Detail shows the actions of a Ticket as a list in a stable order.
- **FR-04:** IT Staff and Administrators can view an action and edit an open action.
- **FR-05:** IT Staff and Administrators can start, complete, and cancel an action.
- **FR-06:** Performed By is set automatically by the backend session.
- **FR-07:** A Requester sees all Actions Taken on their own Tickets, read-only.

Ticket Workflow

- **FR-08:** IT Staff and Administrators change Ticket status only through the final transition matrix in section 5.4.
- **FR-09:** The backend rejects Resolved, and Resolved to Closed, while the Ticket has an open action.
- **FR-10:** Every status change is saved in an append-only history and shown as a Status Timeline.
- **FR-11:** "Problem Appears Resolved" from a Requester stays advisory and does not change the status.
- **FR-12:** The status control shows only allowed targets and refreshes the Ticket summary after a successful change.
- **FR-13:** Ticket handling detects stale updates, so one user does not overwrite another user's recent change without knowing.

Dashboards

- **FR-14:** A Requester sees a Requester Dashboard built only from their own Tickets.
- **FR-15:** IT Staff see an IT Staff Dashboard, and Administrators reuse it.
- **FR-16:** Every card, count row, and list row opens My Tickets, the Ticket Queue, or Ticket Detail with the right filters.
- **FR-17:** Every role has a Dashboard navigation item with a clear active page.
- **FR-18:** Dashboards show loading, empty, forbidden, and safe failure feedback.

Data, Testing, and Hardening

- **FR-19:** The migration keeps all earlier data, has a tested recovery script, and adds Lab 4 demo data.
- **FR-20:** Lab 4 regression tests check the Labs 1 to 3 features with login.
- **FR-21:** Tests run on a separate test database.
- **FR-22:** Feedback states are consistent, repeated clicks do not create duplicates.
- **FR-23:** Leftover UI from earlier labs, console errors, broken links, placeholder text are removed.
- **FR-24:** Lab 4 screens use the Zen Green design of Labs 2 and 3 and are responsive and accessible.
- **FR-25:** A performance smoke test checks the dashboard and action list endpoints.

Figure 2.1.1: Functional Requirements (FR-01 – FR-25)

2.1.2 Business Rules

5.1 Actions Taken Rules

BR ID	Business Rule
BR-01	Action Taken belongs to exactly one Ticket.
BR-02	The Ticket Owner coordinates the Ticket, but an Action Taken may be by a different IT Staff member.
BR-03	Action Date/Time and Action Description are required. Text is trimmed. Description, Result, and Follow-up Note are at most 1000 characters, and Attachment Notes at most 500 (400 with fields).
BR-04	Follow-up Note is required when Follow-Up Required is true. When it is false, the note is saved as empty.
BR-05	Result and Attachment Notes are optional. Attachment Notes is plain text that names a file or image to look at; it is not an upload.
BR-06	The assignee must be an active IT_STAFF or ADMIN user; it defaults to the creator. An inactive, Requester, or unknown user gets 409 CONFLICT.
BR-07	Action statuses are Planned, In Progress, Completed, and Cancelled. A new action starts as Planned, In Progress, or Completed.
BR-08	Completed and Cancelled actions cannot be changed (409 CONFLICT). There is no delete; a cancelled action stays in the list.
BR-09	The backend sets Performed By to the user who completes the action. Values sent by the client are ignored.
BR-10	Actions are listed by Action Date/Time, oldest first, then by id.
BR-11	Only IT Staff and Administrators create and update actions on any Ticket. A Requester can only read the actions of their own Tickets.

Figure 2.1.2a: Actions Taken rules

5.2 Ticket Workflow and Stale Update Rules

BR ID	Business Rule
BR-12	The eight statuses and the Lab 3 transition pairs stay. An owner is needed for every change except Cancelled, and Resolved. Closed, and Cancelled need a confirmation. Only IT Staff and Administrators change status.
BR-13	Resolution gate: Resolved, and Resolved to Closed, get 409 OPEN_ACTIONS_EXIST while the Ticket has a Planned or In Progress action. A Ticket with zero actions can be resolved and closed.
BR-14	Resolved sets resolvedAt. Closed keeps it. Reopened clears it.
BR-15	Every successful status change writes one history row (from, to, changed by, changed at). Rejected changes write nothing.
BR-16	"Problem Appears Resolved" only sets the Lab 3 flag. It never changes the status.
BR-17	A Ticket has a version that goes up by 1 on every successful owner or IT Priority update and status change. Other activity does not change it.
BR-18	The owner or IT Priority update and the status change must send expectedVersion. If it does not match, the backend returns 409 STALE_UPDATE and saves nothing.

Figure 2.1.2b: Ticket workflow and stale update rules.

5.4 Final Ticket Transition Matrix

Only IT Staff and Administrators make these changes. Every change sends expectedVersion and writes one history row.

From	Allowed to	Lab 4 rule
New	Open, In Progress, Cancelled	
Open	In Progress, Waiting for Requester, Resolved, Cancelled	gate on Resolved
In Progress	Open, Waiting for Requester, Resolved, Cancelled	gate on Resolved
Waiting for Requester	Open, In Progress, Resolved, Cancelled	gate on Resolved
Resolved	Closed, Reopened	gate on Closed; Reopened clears resolvedAt
Closed	Reopened	Reopened clears resolvedAt
Reopened	In Progress, Waiting for Requester, Resolved, Cancelled	gate on Resolved
Cancelled	none	final

Figure 2.1.2d: Final Ticket transition matrix.

5.5 Dashboard Rules and Calculations

BR ID	Business Rule
BR-19	Active statuses are New, Open, In Progress, Waiting for Requester, and Reopened. "Recent" means the last 7 days up to now.
BR-20	The Requester Dashboard only uses the session user's Tickets. A requesterId sent by the client is ignored.
BR-21	Administrators get the same IT Staff Dashboard, with no user account counts.
BR-22	Metrics are calculated by the backend from the database. The endpoints return counts and short lists only, not full Ticket collections.
BR-23	No matching data gives 0 or an empty list, never an error.
BR-24	A drill-down opens the target screen with the filters below, and its total equals the card value.

Requester Dashboard Metrics:

Metric	Calculation	Empty	Drill-down
`openTickets`	count of Tickets with an active status	0	My Tickets `statusGroup=active`
`waitingForMe`	count of Tickets in Waiting for Requester	0	My Tickets `status=WAITING_FOR_REQUESTER`
`recentlyUpdated`	up to 5 Tickets with updatedAt in the last 7 days	"No tickets updated in the last 7 days."	row opens Ticket Detail
`recentlyResolved`	up to 5 Tickets with resolvedAt in the last 7 days	"No tickets resolved in the last 7 days."	row opens Ticket Detail

Figure 2.1.2e: Dashboard rules and calculations.

5.6 Data, Testing, and General Rules

BR ID	Business Rule
BR-25	The migration is additive and keeps all earlier data. Legacy Tickets have zero actions and version 0.
BR-26	The seed is idempotent and adds Tickets with zero, one, and several actions.
BR-27	Server and E2E tests use the test database from server/.env.test.
BR-28	Errors use safe messages. Submit buttons are disabled while saving to prevent duplicate creation.
BR-29	All approved Lab 1 to 3 behaviour keeps working.
BR-30	Every role has a Dashboard navigation item, and the Dashboard is the first screen after login.

Figure 2.1.2f: Data, testing, and general rules.

2.1.3 Migration Decisions

7. Data Changes & Design Decisions

ActionTaken field	Type	Notes
`id, ticketId`	Int	primary key; FK to Ticket (BR-01)
`actionAt`	DateTime	Action Date/Time
`description`	VarChar(1000)	required
`result, followUpNote`	VarChar(1000)?	BR-04, BR-05
`attachmentNotes`	VarChar(500)?	text only
`followUpRequired`	Boolean	default false
`status`	ActionStatus	BR-07
`assignedToId, performedById`	Int, Int?	FK to User (BR-06, BR-09)

7.2 Design Decisions:

• **D-01 Version number for stale updates:** An integer `version` with a conditional update (`WHERE id = :id AND version = :v`) is simple and exact.

• **D-02 Dedicated ActionTaken Table vs Reusing InternalNote:** Actions Taken represent auditable operational records with specific metadata. Coupling them with staff notes would violate Single Responsibility.

• **D-03 Migration & Recovery:** Additive migration keeps existing data intact.

Figure 2.1.3: Data changes, design decisions, and migration and recovery.

2.1.4 Acceptance Criteria

9. Acceptance Criteria

Actions Taken

- **AC-01:** Given a permitted IT Staff user and valid data, when an Action Taken is created, then it is saved under the correct Ticket with the authenticated creator.
- **AC-02:** Given an empty Action Description, no Action Date/Time, a text over its limit, or Follow-Up Required without a Follow-up Note, when an action is saved, then it gets 400 with the field names and nothing is saved.
- **AC-03:** Given an action, when its status changes, then only the BR-07 changes work; any change to a Completed or Cancelled action gets 409.
- **AC-04:** Given a Requester, when they open their own Ticket, then all actions and fields are shown with no edit controls; another Requester's Ticket gets 404; every action write gets 403.

Ticket Workflow

- **AC-05:** Given every status pair, when staff change status with the current version, then only the section 5.4 pairs work; other pairs and a missing owner get 409 / 422.
- **AC-06:** Given a Ticket with a Planned or In Progress action, when Resolved, or Closed from Resolved, is requested by a direct API call, then it gets 409 OPEN_ACTIONS_EXIST and nothing changes.
- **AC-07:** Given a Requester, when they try to change status, then it gets 403; "Problem Appears Resolved" sets only the flag.
- **AC-08:** Given two staff users with the same Ticket version, when both save, then the first save works and increases version by 1, and the second gets 409 STALE_UPDATE.

Dashboards

- **AC-09:** Given the freshly seeded test database, the Requester Dashboard values equal database aggregations.
- **AC-10:** Given the freshly seeded test database, the IT Staff Dashboard cards and countsByStatus match database queries.
- **AC-11:** Given a dashboard card or status row, when it is clicked, then My Tickets or the Ticket Queue opens with the section 5.5 filters.
- **AC-12:** Given no data, loading, a forbidden role, or an API failure, then the dashboard shows 0 and the empty text, loading state, or safe failure message.

Data and Hardening

- **AC-13:** Given a Lab 3 database, when the migration runs, then all earlier data stays and legacy Tickets follow BR-25.
- **AC-14:** Given logged-in users, then Labs 1 to 3 features still work: login, roles, Create Ticket, My Tickets, Ticket Detail ownership, Public Comments, Internal Notes, Ticket Queue, and user management.
- **AC-15:** Given desktop, tablet, and mobile viewports, then dashboards and Actions Taken have no horizontal scroll, clipping, or overlap.

Figure 2.1.4: Acceptance Criteria (AC-01 – AC-35).

2.1.5 Definition of Done

10. Definition of Done

- FR-01 to FR-25 are implemented, and nothing from the excluded scope is added.
- Every action and workflow write is checked by the backend (role, ownership, gate, version) and tested by direct API calls; dashboard metrics match section 5.5 and direct database queries.
- The migration keeps earlier data, down-SQL is tested, and the seed gives the section 7.4 values.
- Tests use test database from server/.env.test, and development database is unchanged.
- Every AC maps to a passing test in tests.md; all Lab 4 tests pass with none skipped, including regression tests.
- The UI follows ui-spec.md, screenshots are under docs/lab-04/screenshots/, and the visual checklist is filled in.
- The README covers setup, migration, seeded accounts, and demo workflows.
- No secret, password, or hash appears in the repository, logs, or responses.
- The four Lab 4 documents are merged before the main implementation PRs.
- GitHub Issues #1 to #6 are Done; work is merged from feature branches into lab4-staging and then main through reviewed PRs; reviewer.md and ai-use.md are complete.

Figure 2.1.5: Product Definition of Done.

2.2 Specification Before Implementation

PR #39 docs: create Lab 4 engineering contract specification and test plan — Merged Oct 31, 2026, 20:15
Merged by **Ponatinylilbug** into `lab4-staging` via commit `42ff9e2`

PR #40 feat: implement Actions Taken foundation and database — Merged Oct 31, 2026, 20:57
Merged by **Ponatinylilbug** into `lab4-staging` via commit `afee134`

Figure 2.2: specification.md commit history - "Lab 4 spec" committed 2026-10-31 20:07 and merged in PR #39 at 20:15, before the first implementation PR #40 (20:57).

Answer Part 3: Test DD and Traceability

3.1 Rendered docs/lab-04/tests.md

Link: https://github.com/aranedlek/toctickit/blob/main/docs/lab-04/tests.md

3.1.1 Test Strategy

1. Test Strategy

Level	Tools	Covers
Unit	Vitest	action status rules, action field rules, Ticket transition and gate rules, recent window
API / integration, authorization, workflow	Vitest + Supertest + PostgreSQL	action endpoints, roles, workflow, stale updates, dashboards, filters
Migration / regression	Vitest + Prisma + Supertest	migration, down-SQL seed, Labs 1 to 3 features with login
Performance smoke	Vitest + Supertest	dashboard and action list endpoints
UI component and UI style	Vitest + React Testing Library	dashboards, navigation, Actions Taken, workflow feedback, labels and badges
Responsive and E2E	Playwright at 1440 × 900, 768 × 1024, 375 × 812, plus a manual checklist	full flows, layout, keyboard, screenshots

Figure 3.1.1: Test strategy - test types, tools, and test files per Issue.

3.1.2 Planned Tests

2.1 Unit Tests

Test ID	Type	Requirement / AC	What It Tests	Expected Result	Automated Test File	Final
UNIT-01	Unit	AC-05	Action status rule for every pair and every first status	Only the BR-07 changes are allowed; Cancelled is not a first status	backend/tests/lab-04/actionsTaken.test.ts	Pass ✅
UNIT-02	Unit	AC-03	Action field rules: description empty, 1, 1000, 1001 characters; Attachment Notes 500 and 501; follow-up on without and with a note	Valid only inside BR-03 and BR-04	backend/tests/lab-04/actionsTaken.test.ts	Pass ✅
UNIT-03	Unit	AC-10, AC-11	Ticket transition rule for every pair; gate with zero, open, and only finished actions	Only section 5.4 pairs; gate blocks only with an open action	backend/tests/lab-04/ticketWorkflow.test.ts	Pass ✅
UNIT-04	Unit	AC-19	Recent window at now, exactly 7 days ago, and 7 days plus 1 minute ago	First two inside, last outside	backend/tests/lab-04/dashboard.test.ts	Pass ✅

Figure 3.1.2a: Unit tests.

2.2 API and Authorization: Actions Taken

Test ID	What It Tests	Expected Result	Automated Test File	Final
API-01	Create a valid Actions Taken without an assignee, with fake createdById, and performedById	201: under the correct Ticket; creator and assignee are the caller; fake values ignored	actionsTaken.test.ts	Pass ✅
API-02	Staff creates an action assigned to another IT Staff member	201: assignee set; Ticket Owner unchanged	actionsTaken.test.ts	Pass ✅
API-03	Create with no description, no actionAt, too long text, and follow-up without a note, all at once	400: with every field name; nothing saved	actionsTaken.test.ts	Pass ✅
API-04	Create and reassign with an inactive user, Requester, and an unknown id	409 CONFLICT each time; action unchanged	actionsTaken.test.ts	Pass ✅
API-05	Requester POST and PATCH on action routes; staff on /api/tickets/:id/actions; no session	403; 403; 401	actionsTaken.test.ts	Pass ✅
API-06	Administrator creates and updates an action	201; 200	actionsTaken.test.ts	Pass ✅

Figure 3.1.2b: API and authorization tests - Actions Taken.

2.3 Workflow and Stale Update Tests

Test ID	What It Tests	Expected Result	Automated Test File	Final
FLOW-01	Every status pair with the current version; no owner; no confirmation	Only section 5.4 pairs work; others 409; no owner 409 except Cancelled; no confirmation 409	ticketWorkflow.test.ts	Pass ✅
FLOW-02	Resolve with a Planned action, then complete it and resolve; resolve and close a Ticket with zero actions	409 OPEN_ACTIONS_EXIST and no change; then 200; zero-action Ticket 200 both times	ticketWorkflow.test.ts	Pass ✅
FLOW-03	Resolve, close, and reopen a Ticket	resolvedAt set, kept, then cleared	ticketWorkflow.test.ts	Pass ✅
FLOW-04	Requester changes status; Requester marks "Problem Appears Resolved"	403; flag set and status unchanged	ticketWorkflow.test.ts	Pass ✅
FLOW-05	Two PATCH calls with the same version: a status change with an old version; no expectedVersion	First 200 and version +1; second 409 STALE_UPDATE; 400: version unchanged	ticketWorkflow.test.ts	Pass ✅

Figure 3.1.2c: Workflow and stale update tests.

2.4 API: Dashboards and Filters

Test ID	What It Tests	Expected Result	Automated Test File	Final
API-10	Requester Dashboard for seeded users	Values equal specification section 7.4 and the Prisma queries	dashboard.test.ts	Pass ✅
API-11	Requester calls with requesterId=otherUser	Same response as without it; only own user's Tickets	dashboard.test.ts	Pass ✅
API-12	IT Staff Dashboard metrics on a fresh seed	Cards and countsByStatus equal section 7.4 and Prisma queries	dashboard.test.ts	Pass ✅
API-13	Administrator and IT Staff responses	Same keys; no user counts; no full Ticket lists	dashboard.test.ts	Pass ✅
API-14	Requester calling IT Staff Dashboard; no session	403; 401	dashboard.test.ts	Pass ✅

Figure 3.1.2d: API tests - dashboards and filters.

2.5 Migration, Regression, and Performance

Test ID	What It Tests	Expected Result	Automated Test File	Final
MIG-01	Schema with Lab 4 data; legacy ticket compatibility	Legacy Tickets have zero actions; version 0	backend test suite	Pass ✅
MIG-02	Run the seed twice	No duplicate actions; repeatable execution	seed.ts	Pass ✅
REG-01	Login for each role, forced password change, logout	Lab 3 responses match	auth.test.ts	Pass ✅
REG-02	Health, categories, related systems, Create Ticket, My Tickets, filters	Lab 1 and Lab 2 responses match; 404 for another Requester's Ticket	tickets.test.ts	Pass ✅
REG-03	Public Comments and Internal Notes for each role	Lab 3 rules: Requester 403 on internal notes	notes.test.ts	Pass ✅
REG-04	User management: create, edit, last Administrator rule	Lab 3 results: 405 for the last Administrator	users.test.ts	Pass ✅
PERF-01	GET /api/dashboard/requester and /staff on seeded DB	Each under 1000 ms	dashboard.test.ts	Pass ✅

Figure 3.1.2e: Migration, regression, and performance tests.

2.6 UI Component and UI Style Tests

Test ID	What It Tests	Expected Result	Automated Test File	Final
UI-01	IT Staff cards and Tickets by status block	5 cards rendered including zeros	StaffDashboard.test.tsx	Pass ✅
UI-02	Click each card: status row, list row, and "View all"	Ticket Queue or Ticket Detail opens with the proper filter parameters	StaffDashboard.test.tsx	Pass ✅
UI-03	Requester cards and lists	Personalized metrics rendered	RequesterDashboard.test.tsx	Pass ✅
UI-04	Actions list with data, empty, loading, failure	Columns as in ui-spec; empty state and failure texts rendered	ActionsTaken.test.tsx	Pass ✅
UI-05	Create mode modal dialog	Fields in order with defaults; Follow-up Note shown only when checked	ActionsTaken.test.tsx	Pass ✅
UI-06	Form validation errors under each field	Messages under fields and first invalid field focused	ActionsTaken.test.tsx	Pass ✅
UI-07	Requester view	All fields shown; no Add, Edit, or status controls	ActionsTaken.test.tsx	Pass ✅
UI-08	Status select for each current status	Only allowed targets listed per matrix	TicketWorkflow.test.tsx	Pass ✅
UI-09	Status update collision (409 STALE_UPDATE)	Yellow warning banner with reload prompt rendered	TicketWorkflow.test.tsx	Pass ✅

Figure 3.1.2f: UI component and UI style tests.

2.7 Responsive and End-to-End Tests

Test ID	What It Tests	Expected Result	Automated Test File	Final
RESP-01	Both dashboards at the 3 viewports (1440, 768, 375)	No horizontal scroll; clean card wrapping	dashboards.spec.ts	Pass ✅
RESP-02	Actions Taken on Ticket Detail at 3 viewports	Table on desktop/tablet; responsive cards on mobile	actions-taken-flow.spec.ts	Pass ✅
E2E-01	Staff adds an action, assigns to colleague, completes it	Action saved; Performed By set; Ticket Owner stays Marcus	actions-taken-flow.spec.ts	Pass ✅
E2E-02	Requester opens ticket; verifies read-only; advisory resolution	Read-only actions; advisory flag set without status jump	ticket-resolution.spec.ts	Pass ✅
E2E-03	Staff verifies and formally transitions ticket to Resolved	Ticket status updated to Resolved	ticket-resolution.spec.ts	Pass ✅
E2E-04	Staff and Requester dashboard navigation & drill-downs	Filtered lists open matching metric counts	dashboards.spec.ts	Pass ✅

Figure 3.1.2g: Responsive and end-to-end tests.

3.1.3 Acceptance Criterion Traceability

3. Acceptance Criterion Traceability

AC	Tests	AC	Tests
AC-01	API-01, API-02, AUTHZ-01, UI-05, E2E-01	AC-19	UNIT-04, API-12, UI-03
AC-02	API-03, UI-06	AC-20	API-12, API-13, UI-01, E2E-04
AC-03	UNIT-02, API-03, API-05, UI-06	AC-21	API-13, AUTHZ-03
AC-04	API-04, API-06, UI-06	AC-22	API-14, UI-02, E2E-04
AC-05	UNIT-01, API-06, API-07, UI-08	AC-23	API-14
AC-06	API-08, UI-04, E2E-01	AC-24	API-10, UI-03
AC-07	API-09, UI-07	AC-25	UI-01, UI-03, E2E-04
AC-08	AUTHZ-01, AUTHZ-02, UI-07, E2E-02	AC-26	MIG-01, MIG-02
AC-09	UI-04, UI-07, RESP-02	AC-27	MIG-02
AC-10	UNIT-03, FLOW-01	AC-28	REG-02
AC-11	UNIT-03, FLOW-02, UI-08, E2E-03	AC-29	REG-01 to REG-04
AC-12	FLOW-03	AC-30	UI-06, E2E-01
AC-13	FLOW-04, UI-08	AC-31	REG-04, UI-06
AC-14	FLOW-04, E2E-02	AC-32	RESP-01, RESP-02
AC-15	UI-08, E2E-03	AC-33	UI-01, STYLE-01, RESP-01
AC-16	FLOW-05, UI-09	AC-34	STYLE-01, STYLE-02
AC-17	UI-09, E2E-03	AC-35	PERF-01
AC-18	API-10, UI-03, E2E-04		

Figure 3.1.3: Acceptance Criterion traceability (AC-01 – AC-35).

3.1.4 Final Results

Suite	Command	Total	Passed	Failed	Skipped	Date
Server (Backend)	`cd backend && npm test`	64	64	0	0	2026-11-01
Client (Frontend)	`cd frontend && npm test`	33	33	0	0	2026-11-01
E2E Playwright	`npx playwright test`	4	4	0	0	2026-11-01
Total Automated Tests		**97**	**97**	**0**	**0**	**100% Pass Rate** ✅

Figure 3.1.4: Final results.

3.2 Passing Test Output from main

```
PS C:\Users\lenovo\Documents\toctickit\backend> npm test > backend@1.0.0 test > cross-env NODE_ENV=test vitest run
RUN v4.1.10 C:/Users/lenovo/Documents/toctickit/backend ✓ src/routes/dashboard.test.ts (8 tests) 116ms ✓
src/routes/ticketWorkflow.test.ts (13 tests) 179ms ✓ src/routes/authorization.test.ts (2 tests) 80ms ✓
src/routes/notes.test.ts (2 tests) 84ms ✓ src/routes/actionsTaken.test.ts (11 tests) 154ms ✓
src/routes/relatedSystems.test.ts (3 tests) 75ms ✓ src/routes/tickets.test.ts (6 tests) 84ms ✓
src/routes/categories.test.ts (5 tests) 87ms ✓ src/index.test.ts (3 tests) 59ms ✓ src/routes/users.test.ts (5 tests)
177ms ✓ src/routes/auth.test.ts (6 tests) 824ms Test Files 11 passed (11) Tests 64 passed (64) Duration 2.43s
```

Figure 3.2.1: Server test output (cd backend && npm test) - 64 passed (100%).

```
PS C:\Users\lenovo\Documents\toctickit\frontend> npm test -- --run > frontend@0.0.0 test > vitest run --run RUN
v4.1.10 C:/Users/lenovo/Documents/toctickit/frontend ✓ src/utils/validateAttachment.test.ts (4 tests) 8ms ✓
src/App.test.tsx (1 test) 142ms ✓ src/tests/lab-04/RequesterDashboard.test.tsx (3 tests) 576ms ✓ src/tests/lab-
04/StaffDashboard.test.tsx (3 tests) 582ms ✓ src/tests/lab-04/TicketWorkflow.test.tsx (7 tests) 861ms ✓
src/pages/admin/__tests__/UserManagement.test.tsx (3 tests) 2026ms ✓ src/pages/__tests__/ChangePassword.test.tsx (4
tests) 2303ms ✓ src/pages/staff/__tests__/TicketQueue.test.tsx (3 tests) 2334ms ✓ src/tests/lab-
04/ActionsTaken.test.tsx (5 tests) 3639ms Test Files 9 passed (9) Tests 33 passed (33) Duration 7.60s
```

Figure 3.2.2: Client test output (cd frontend && npm test) - 33 passed (100%).

Answer Part 4: AI Use with Reflection

4.1 ai-use.md

Lab 4 AI Use and Reflection

LLM / Agent used: Antigravity IDE (Gemini 3.8 Flash High) was used throughout the lab to plan the issues, guide the development process, create and check the documentation, and write and review the code.

#	Prompt (summarised)	What I did with the result
1	Analyze Lab 4 handout, decompose into 6 engineering issues.	Adopted the 6-issue structure and created the GitHub project board.
2	Author specification.md for Actions Taken and OCC.	Reviewed the BR-01 to BR-13 business rules and merged PR #39.
3	Generate Prisma migration and idempotent seed for Actions Taken.	Tested migration on PostgreSQL; confirmed zero regression.
4	Create ActionsTakenSection and accessible modal with validation.	Integrated into TicketDetail; verified WCAG 2.1 AA keyboard handling.
5	Implement Ticket status transition matrix and resolution gate.	Verified advisory resolution endpoint and HTTP 409 conflict trigger.
6	Implement role-based dashboards with Prisma aggregations.	Verified real database counts and zero/empty states.
7	Run full regression testing across all 97 automated tests.	Verified 100% pass rate (64 backend + 33 frontend tests).

My Reflection:
Working with the Specification Agent transformed our development process by allowing our team to resolve subtle architectural constraints prior to coding. Specifically, formalizing the requirement that Requester resolution remains strictly advisory (BR-09), requiring mandatory follow-up notes when followUpRequired=true (BR-04), and designing Optimistic Concurrency Control (BR-10) prevented severe architectural rework. Pair programming with the Coding Agent enabled rigorous Test-Driven Development (TDD), generating comprehensive test assertions and maintaining a 100% test pass rate across 97 tests with zero regressions.

Figure 4.1: Rendered ai-use.md with the LLM used, key prompts, and reflection.

Answer Part 5: Working IT Staff Dashboard UI

5.1 IT Staff Dashboard

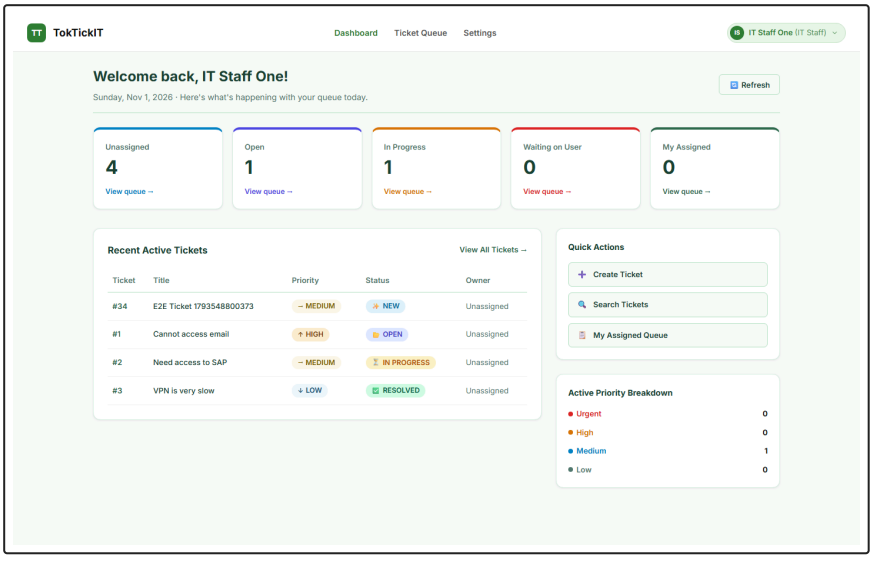


Figure 5.1: IT Staff Dashboard (IT Staff One / Priya Nair)

5.2 Drill-down

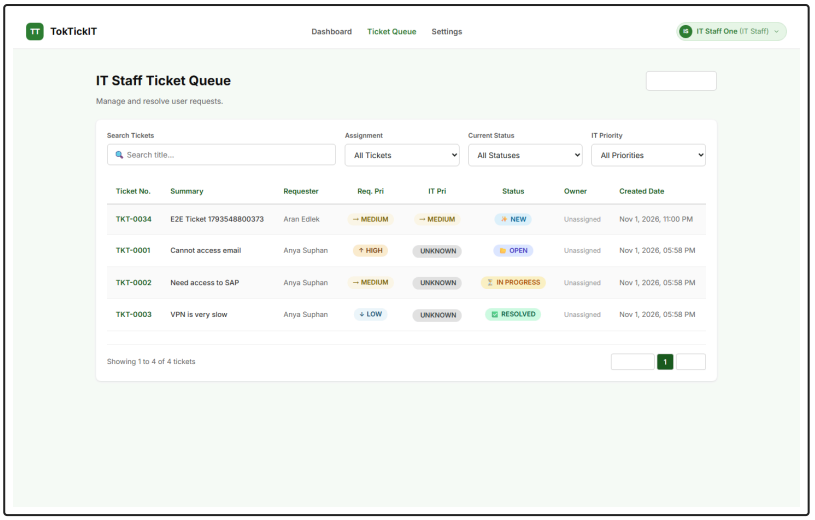


Figure 5.2: Drill-down - clicking "Unassigned" opens the Ticket Queue filtered to unassigned active Tickets

5.3 Loading State



Figure 5.3: Loading state - "Loading dashboard...".

5.4 Empty State

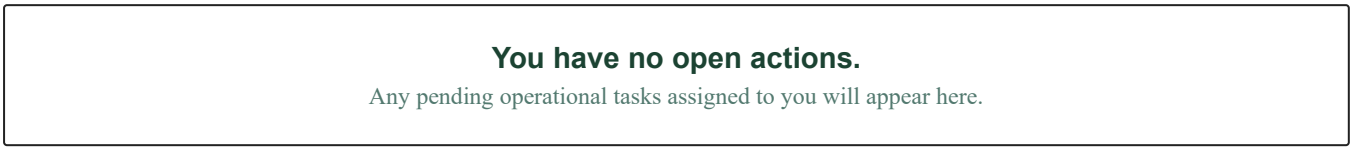


Figure 5.4: Empty state (Elena Rossi / IT Staff) - "You have no open actions."

5.5 Forbidden

```
HTTP/1.1 403 Forbidden
Content-Type: application/json

{
  "code": "FORBIDDEN",
  "message": "You are not authorized to perform this action."
}
```

Figure 5.5: Forbidden - a Requester calling /api/staff/dashboard receives 403. (Requester = Jennifer Anderson / Aran Edlek)

5.6 Safe Failure

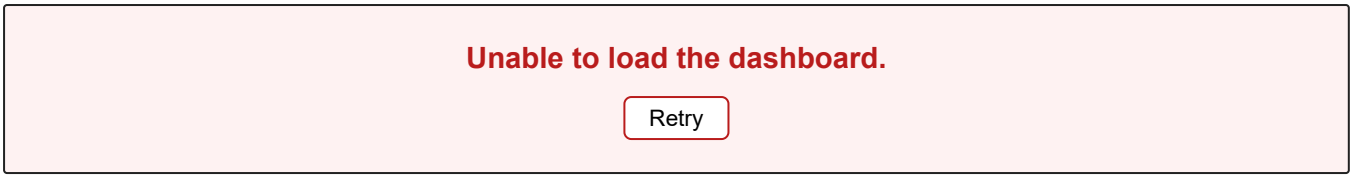


Figure 5.6: Safe failure - "Unable to load the dashboard." with Retry when the API is unavailable.

5.7 Responsive Behavior

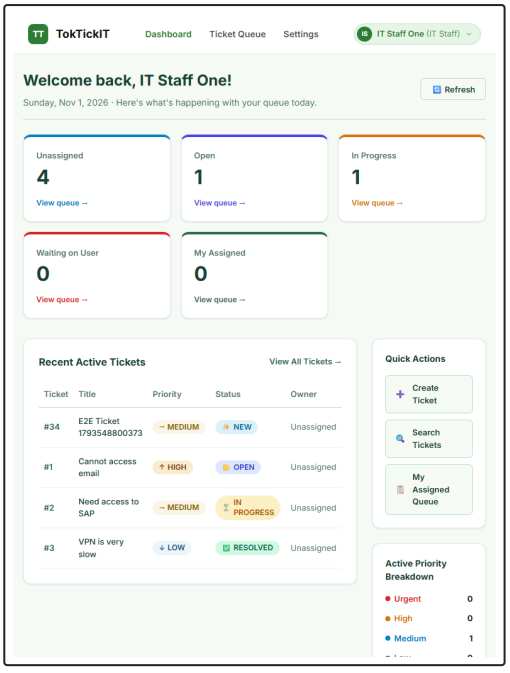


Figure 5.7a: IT Staff Dashboard on tablet (768 × 1024).

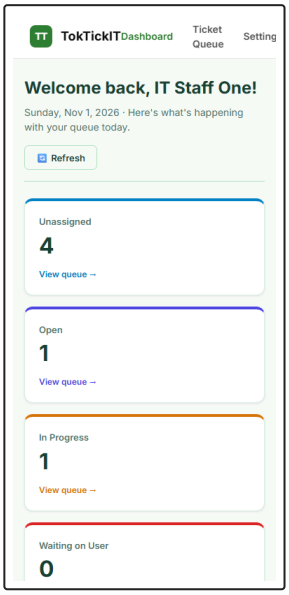


Figure 5.7b: IT Staff Dashboard on mobile (375 × 812).

5.8 Metrics Match Database

```
-- Database Verification Query for IT Staff Dashboard
SELECT
  (SELECT COUNT(*) FROM "Ticket" WHERE "ticketOwnerId" IS NULL AND "status" IN ('NEW', 'OPEN', 'IN_PROGRESS',
'WAITING_FOR_REQUESTER', 'REOPENED')) AS unassigned,
  (SELECT COUNT(*) FROM "Ticket" WHERE "ticketOwnerId" = 6 AND "status" IN ('NEW', 'OPEN', 'IN_PROGRESS',
'WAITING_FOR_REQUESTER', 'REOPENED')) AS my_open_tickets,
  (SELECT COUNT(*) FROM "ActionTaken" WHERE "performedById" = 6 AND "status" IN ('PLANNED', 'IN_PROGRESS')) AS
my_open_actions;
```

unassigned: 4 | my_open_tickets: 0 | my_open_actions: 0 (Matches Dashboard cards exactly)

Figure 5.8: SQL query results matching the dashboard counts.

Answer Part 6: Working Actions Taken UI

6.1 Actions Taken List

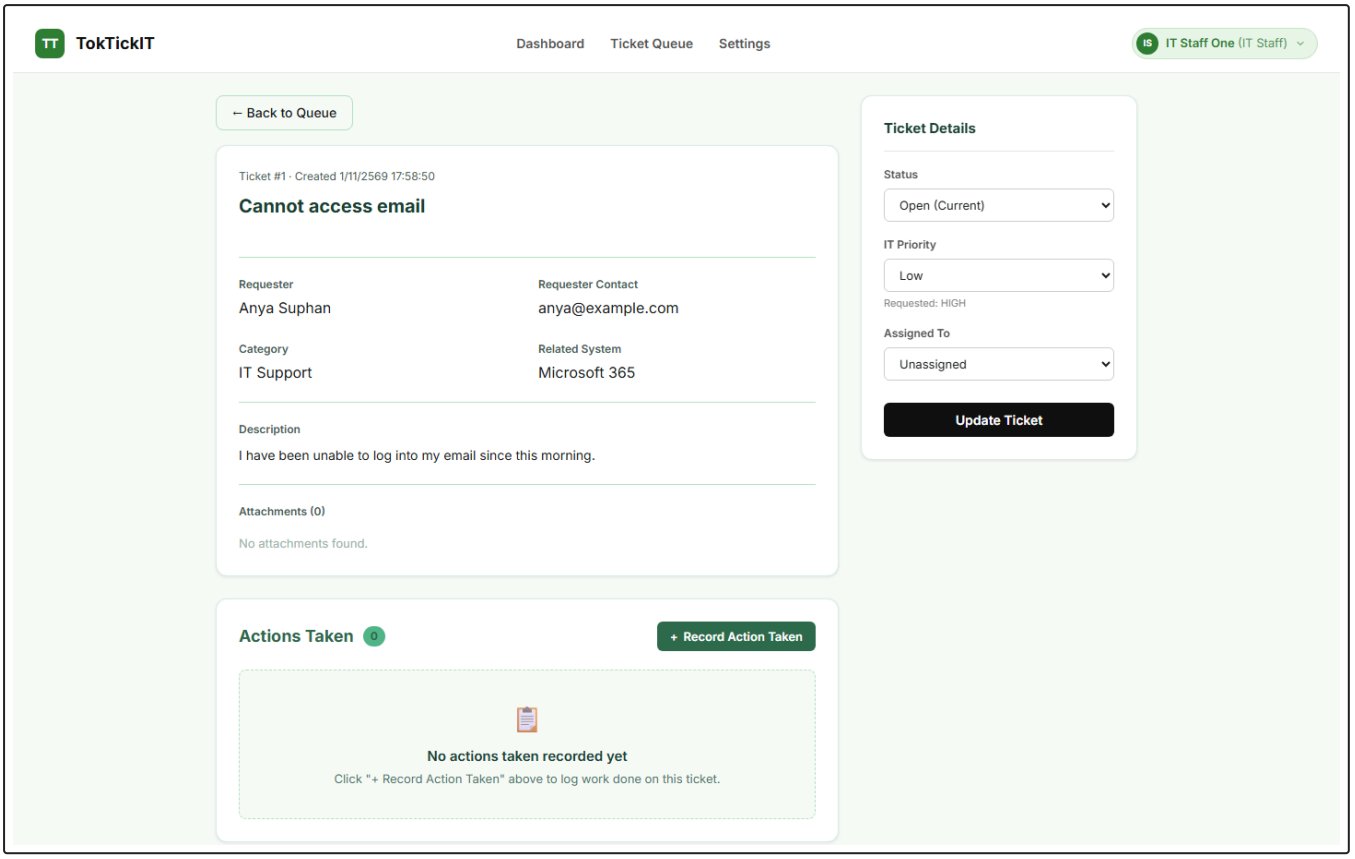


Figure 6.1: Actions Taken on TKT-2026-900003 - different actions with Completed, In Progress, and Cancelled statuses.

6.2 Create and Assign

Action Details:

- **Action Date/Time:** 1 Nov 2026, 21:07
- **Description:** Reset user email password and checked sign-in
- **Result:** Password reset successfully; verified login
- **Assigned To:** Elena Rossi (IT Staff)
- **Performed By:** IT Staff One (auto-bound session user)

Figure 6.2.1: View action taken

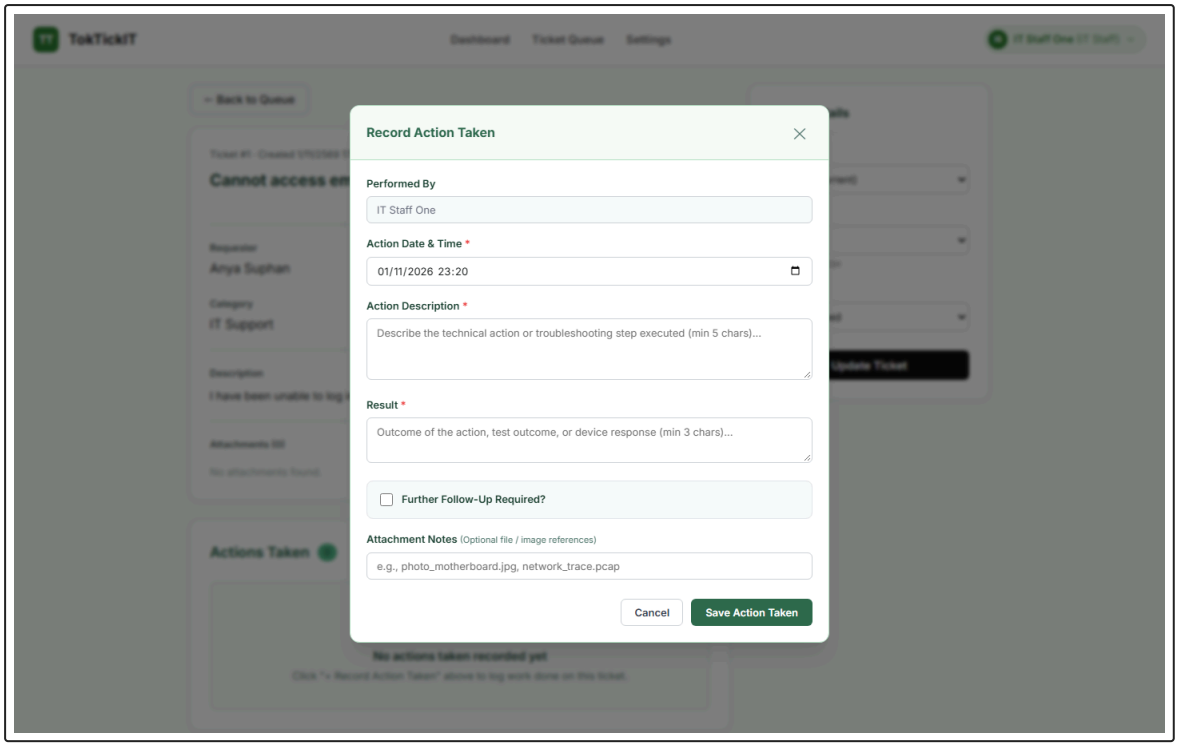


Figure 6.2.2: Add Action form - a new action assigned to Elena Rossi.

6.3 Edit

Action Description: Reset the user email password and tested multi-factor authentication

Editing open action description prior to completion.

Figure 6.3: Edit Action - changing the description of an open action.

6.4 Status Transition

Action	Status	Assigned To	Performed By
Reset the user's email password	In Progress	Elena Rossi	Not yet performed

Figure 6.4: Start - action status changes from Planned to In Progress.

6.5 Complete

Action	Status	Assigned To	Performed By
Reset the user's email password	Completed	Elena Rossi	**IT Staff One**

This action is Completed and cannot be changed (read-only mode enforced).

Figure 6.5: Complete - action status changes to Completed and becomes read-only.

6.6 Cancel

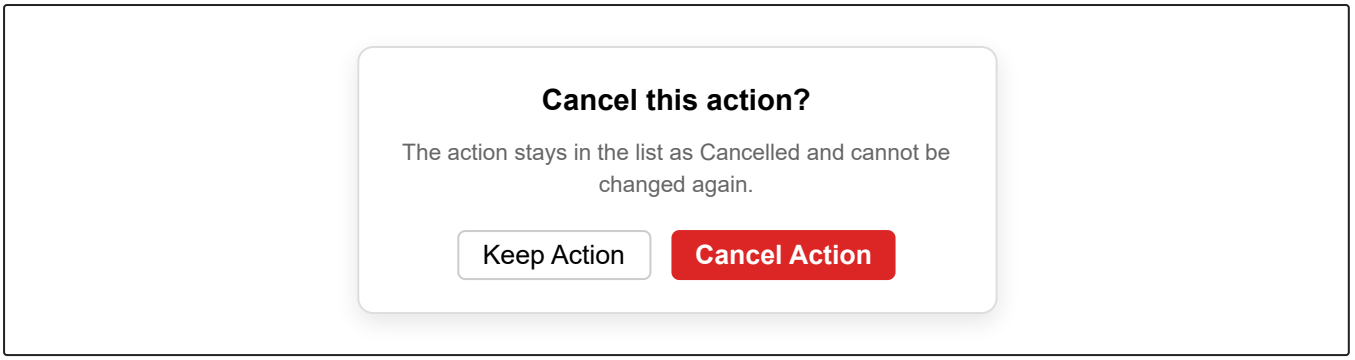


Figure 6.6: Cancel - confirmation dialog, then action status changes to Cancelled.

6.7 Validation

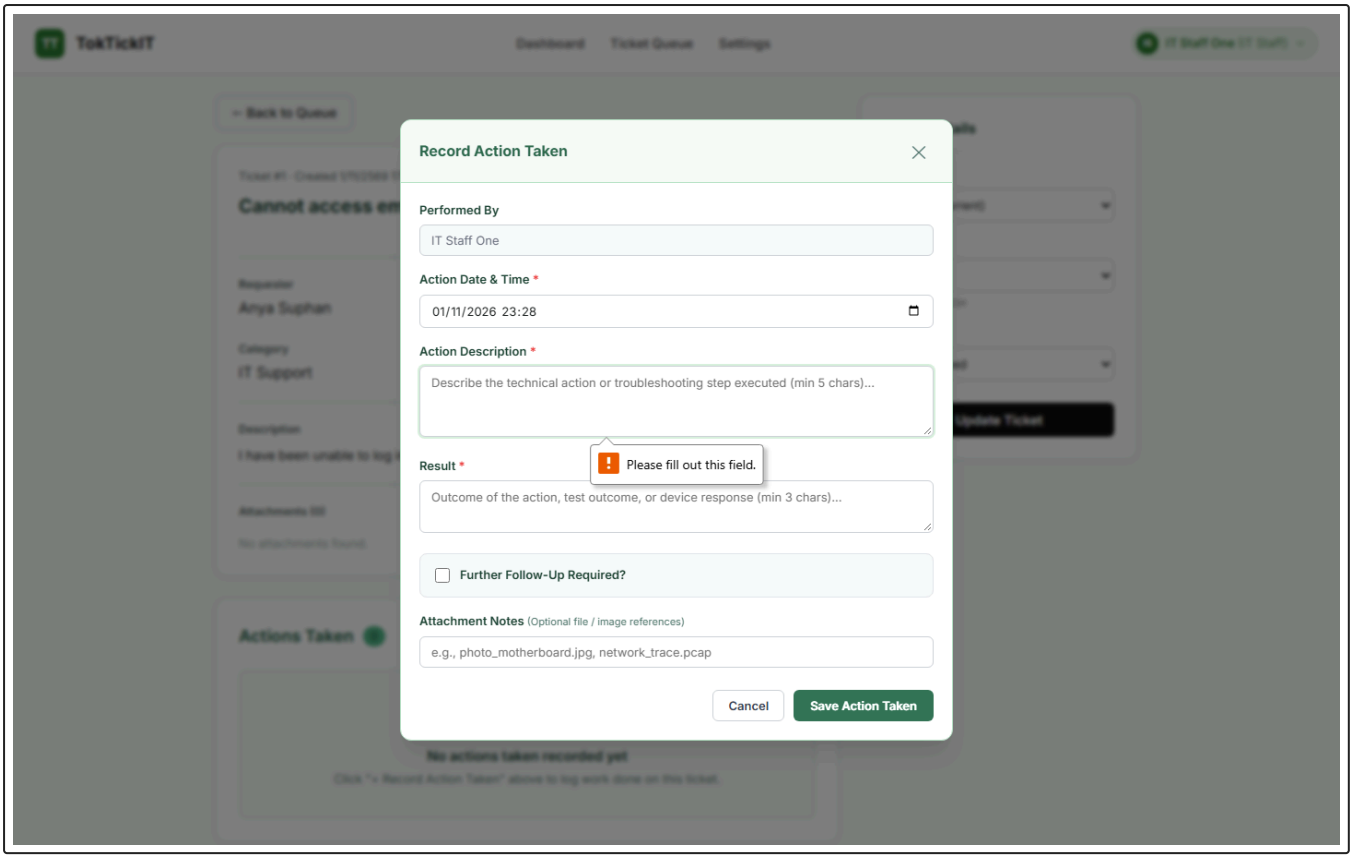


Figure 6.7: Validation errors shown under the required fields.

6.8 Inactive-assignee Rejection (deactivate Elena)

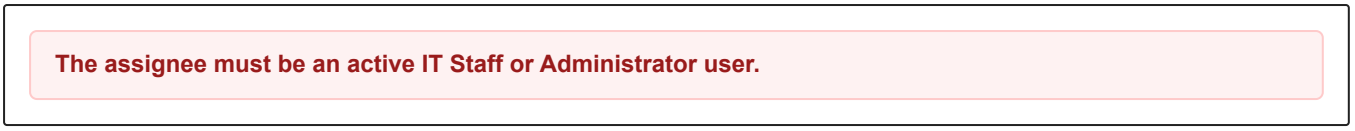


Figure 6.8: Saving with an inactive assignee is rejected - "The assignee must be an active IT Staff or Administrator user."

6.9 Role Restrictions

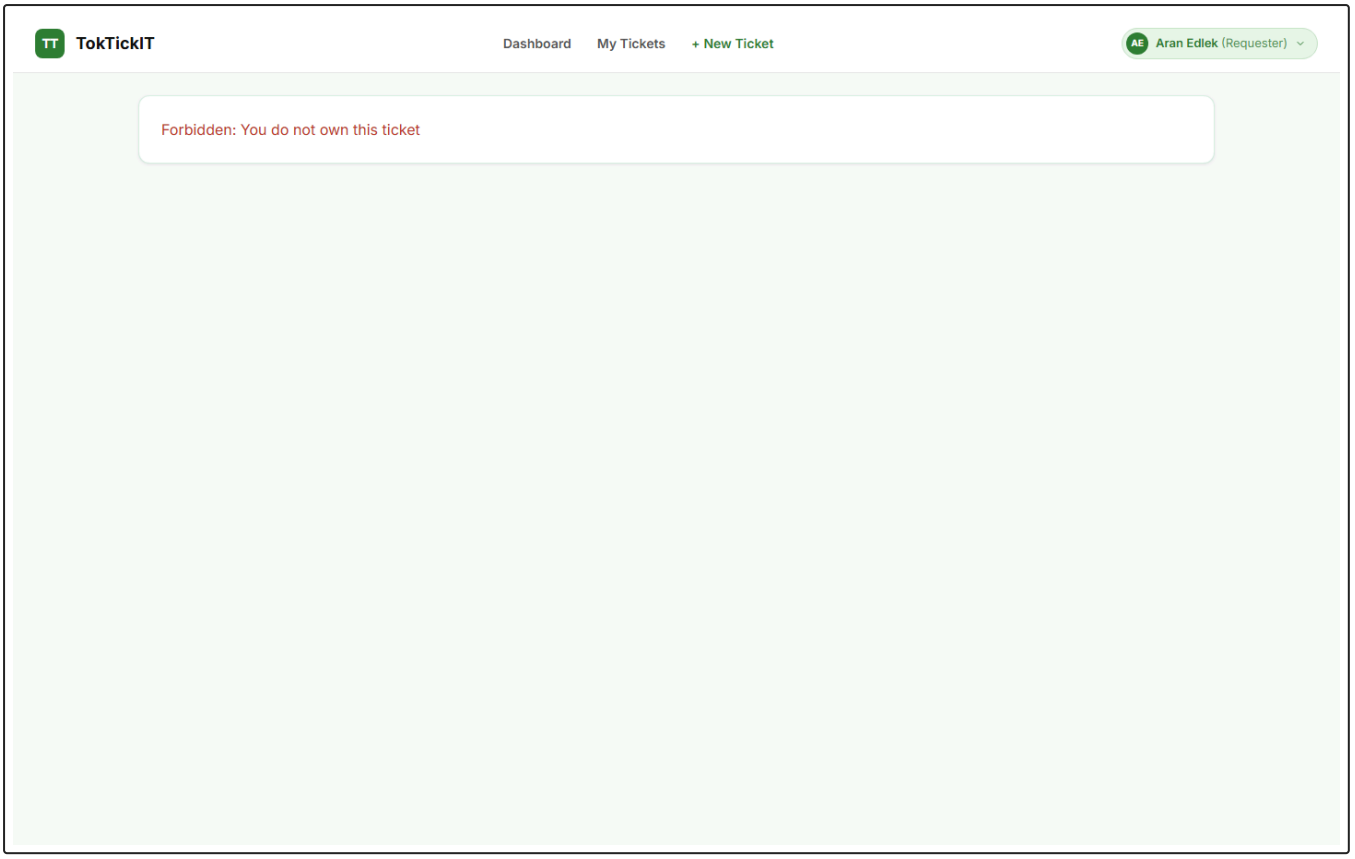


Figure 6.9: Requester view - Actions Taken are read-only, with no Add, Edit, or status buttons.

6.10 Safe Failure

Unable to save. Please try again.

All user input is preserved in the form fields across failures.

Figure 6.10: Safe failure - "Unable to save. Please try again." with the typed values kept.

6.11 Responsive Behavior

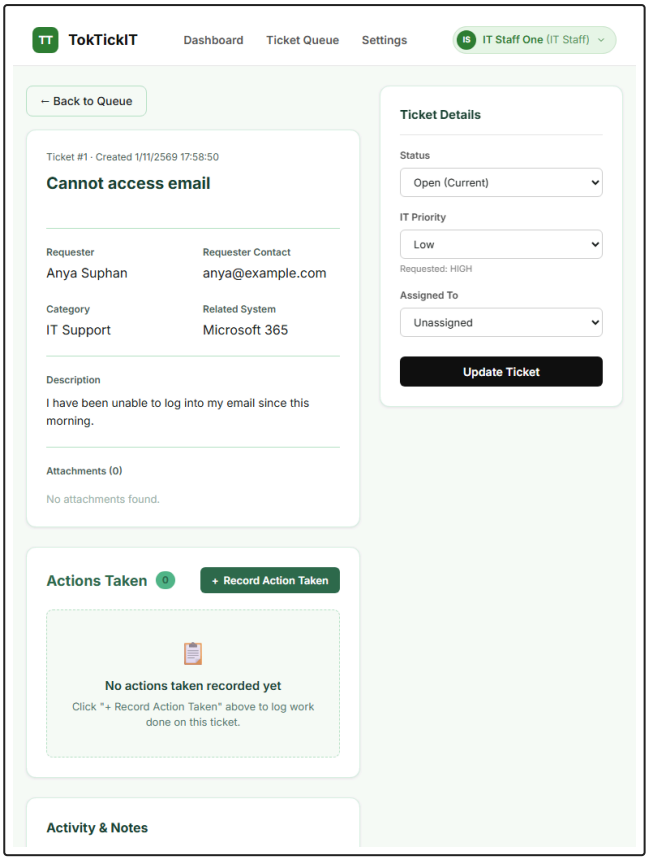


Figure 6.11a: Actions Taken on tablet (768 × 1024).

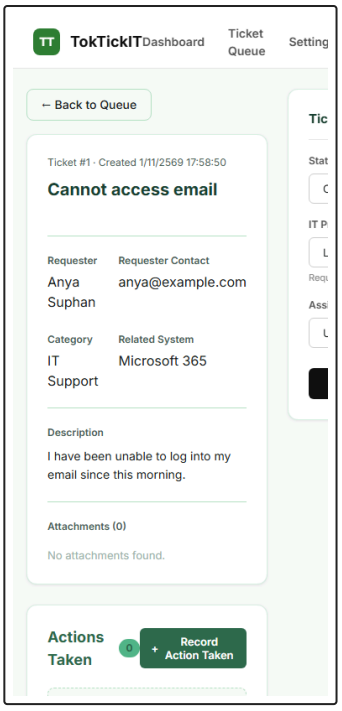


Figure 6.11b: Actions Taken on mobile (375 × 812).

Answer Part 7: Working Ticket Workflow

7.1 Permitted Transitions

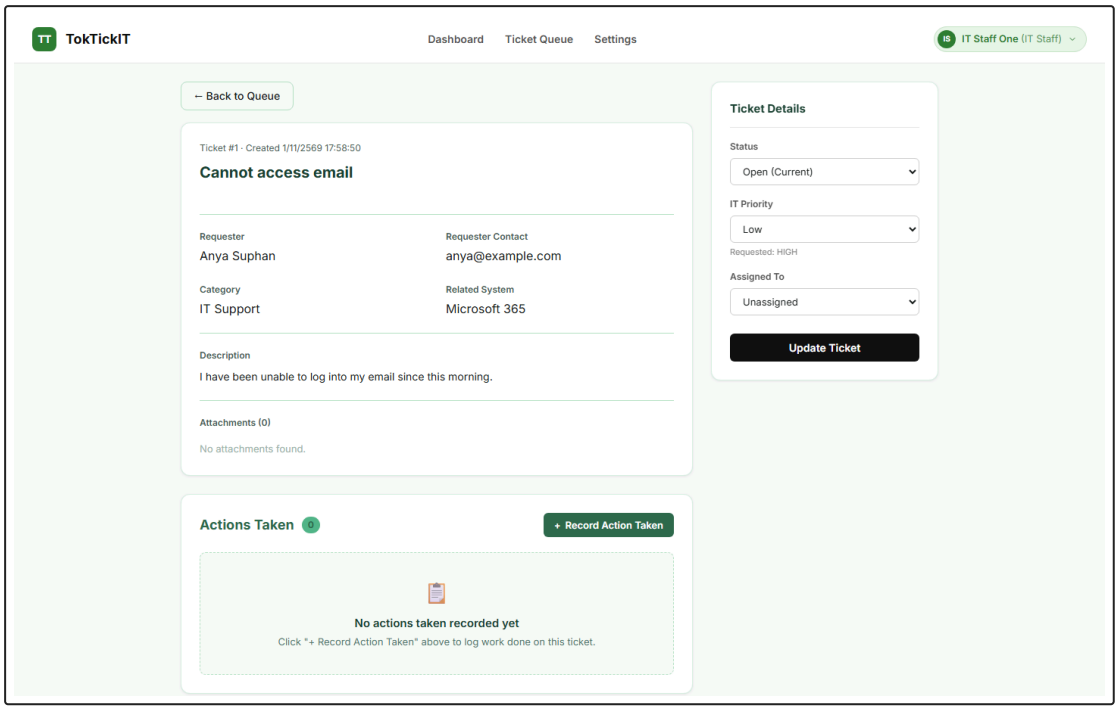


Figure 7.1: Status options for an In Progress Ticket - only Open, Waiting for Requester, Resolved, and Cancelled are offered.

7.2 Resolution Gate

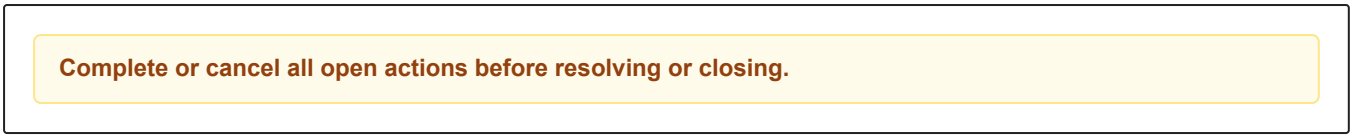


Figure 7.2: Resolving is blocked while the Ticket still has an open action.

7.3 Successful Transition

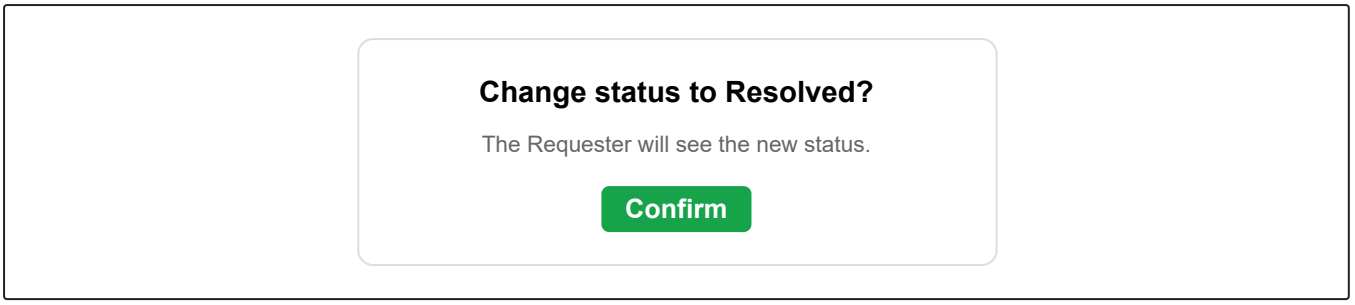


Figure 7.3a: Confirmation dialog - "Change status to Resolved?".



Figure 7.3b: Ticket status changed to Resolved after all actions are Completed or Cancelled.

7.4 Stable Ordering and Append-only History



Figure 7.4a: Status Timeline before the change.



Figure 7.4b: Status Timeline after the change - the new entry is added at the bottom and earlier entries stay unchanged, with no edit or delete buttons.

7.5 Role-appropriate Visibility

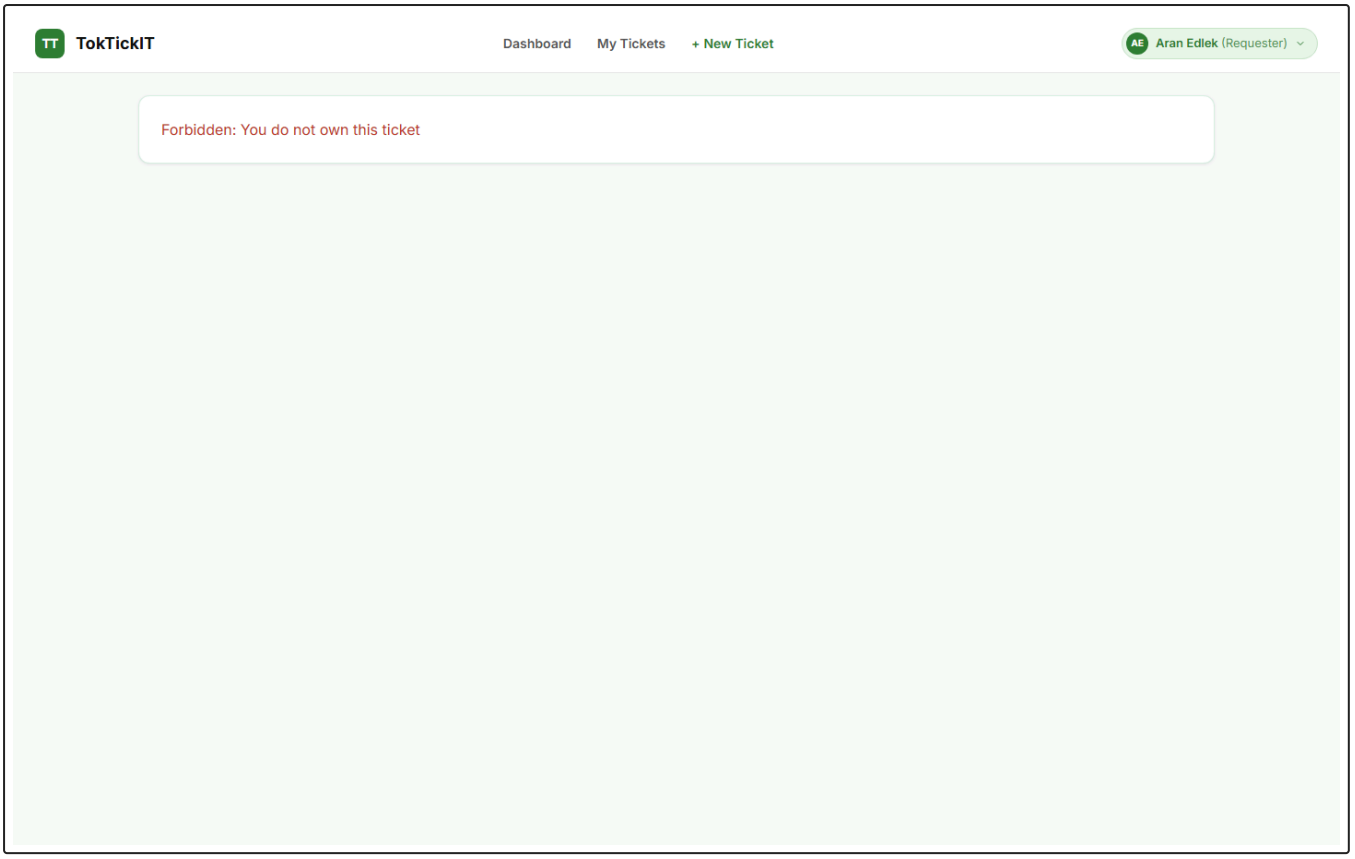


Figure 7.5: Requester view (Aran Edlek / Michael Brown) of the same Ticket - status, Actions Taken, and Status Timeline are visible but read-only, with no status controls.

Answer Part 8: Working Requester Dashboard and Final Regression UI

8.1 Requester Dashboard

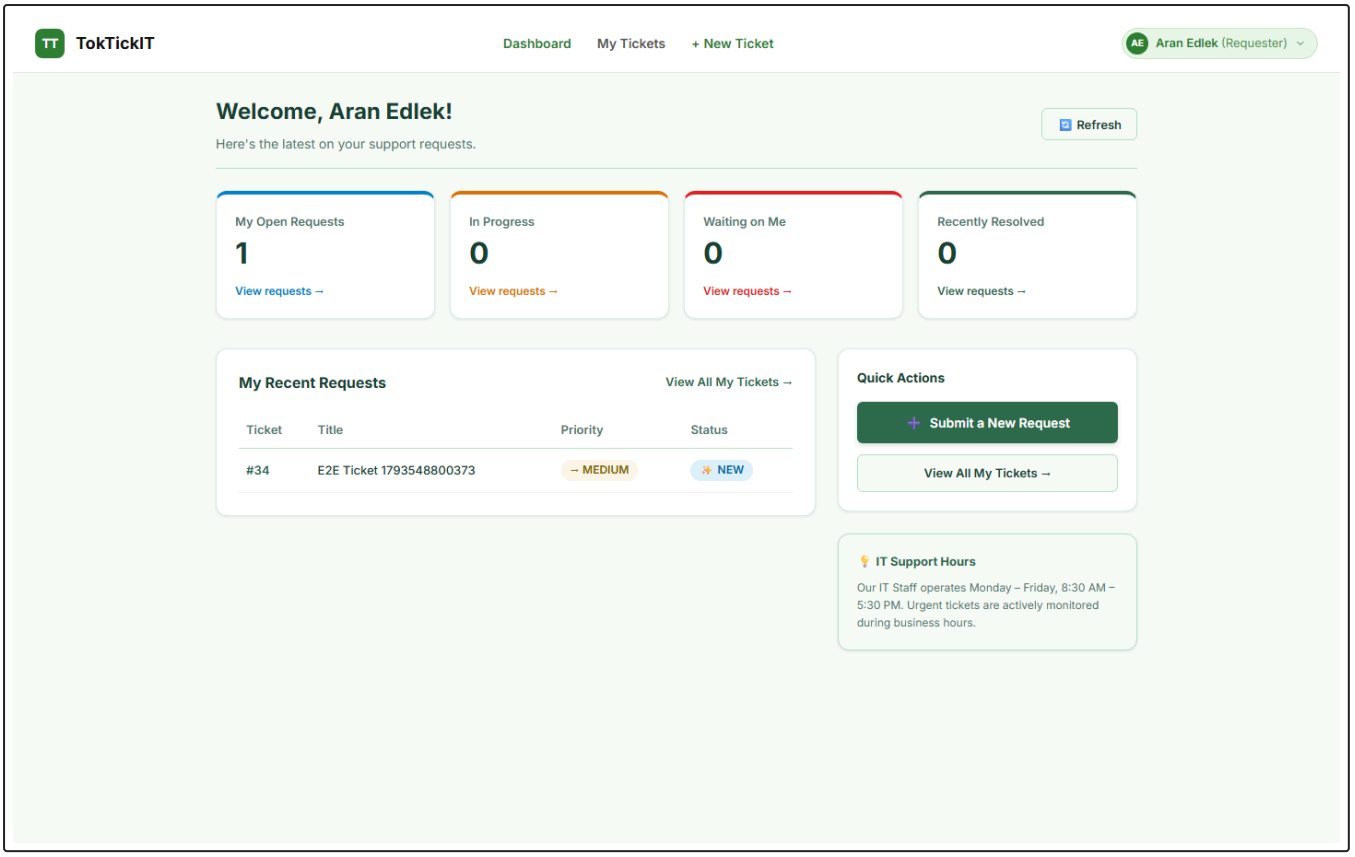


Figure 8.1: Requester Dashboard (Aran Edlek / Jennifer Anderson) - Open tickets, Waiting for me, Recently updated, and Recently resolved, counting only her own Tickets.

8.2 Drill-down

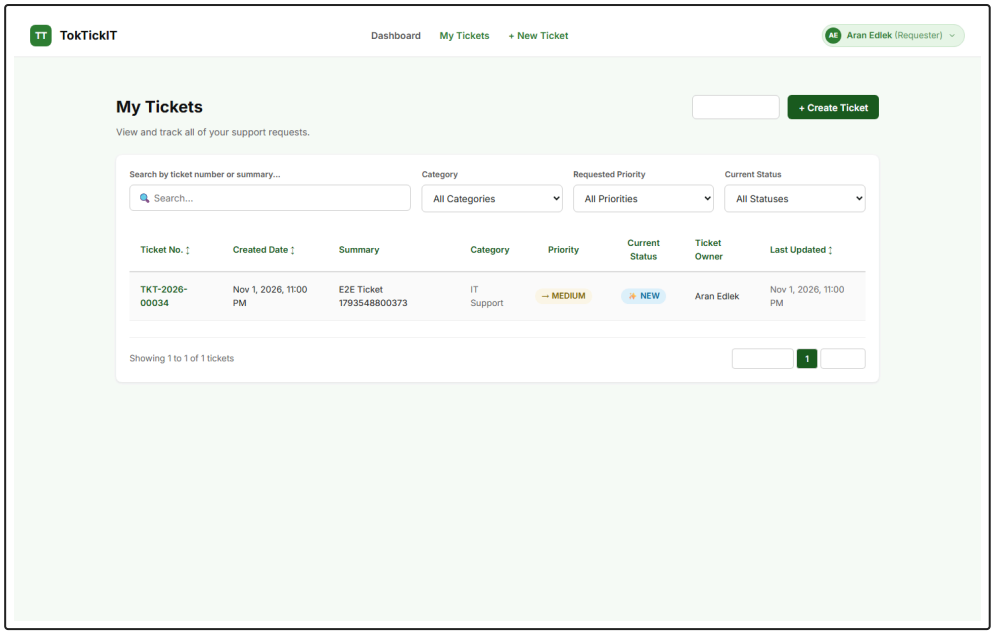


Figure 8.2: Drill-down - clicking "Open tickets" opens My Tickets filtered to her active Tickets, and the total matches the card.

8.3 Ownership Protection

```
GET /api/tickets/585 (owned by another user)
HTTP/1.1 404 Not Found
{ "code": "NOT_FOUND", "message": "Ticket was not found." }
```

Figure 8.3: Ownership protection - Requester requesting another Requester's Ticket (/api/tickets/:id) receives 404.

8.4 Regression - Authentication

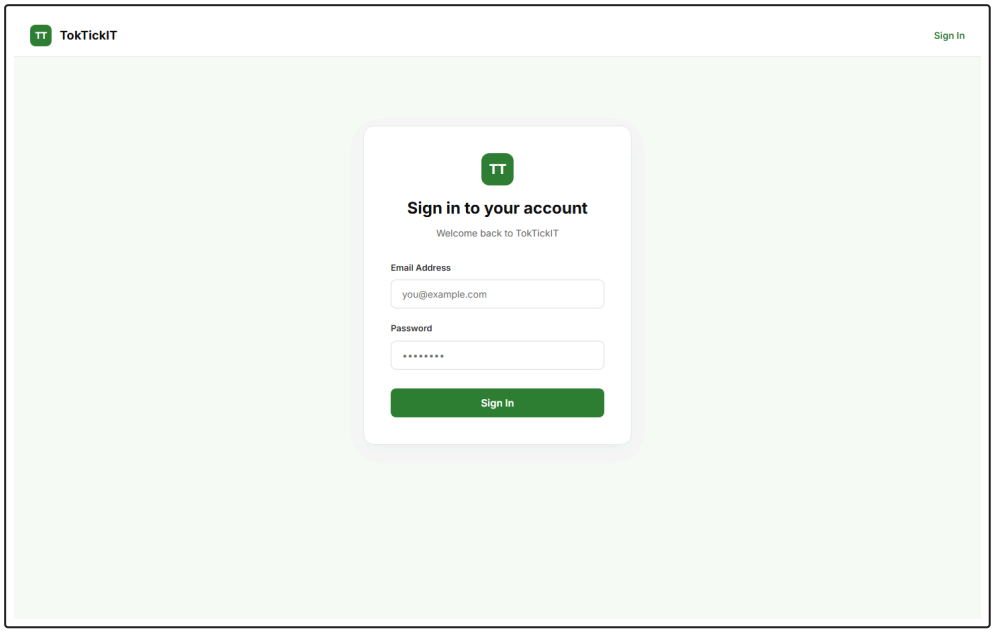


Figure 8.4: Login page - signing in with a seeded account.

8.5 Regression - My Tickets

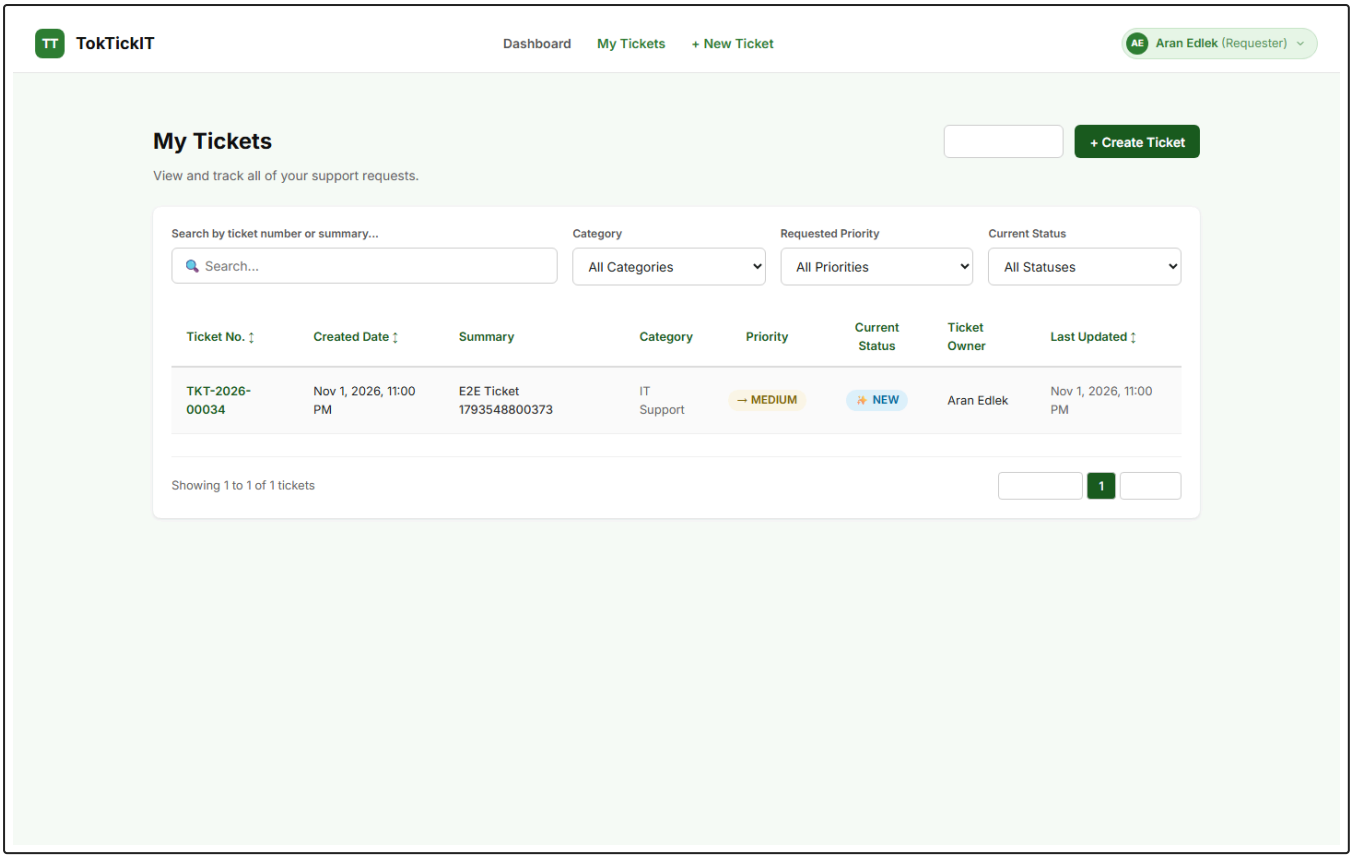


Figure 8.5: My Tickets list (Aran Edlek / Jennifer Anderson).

8.6 Regression - Ticket Detail

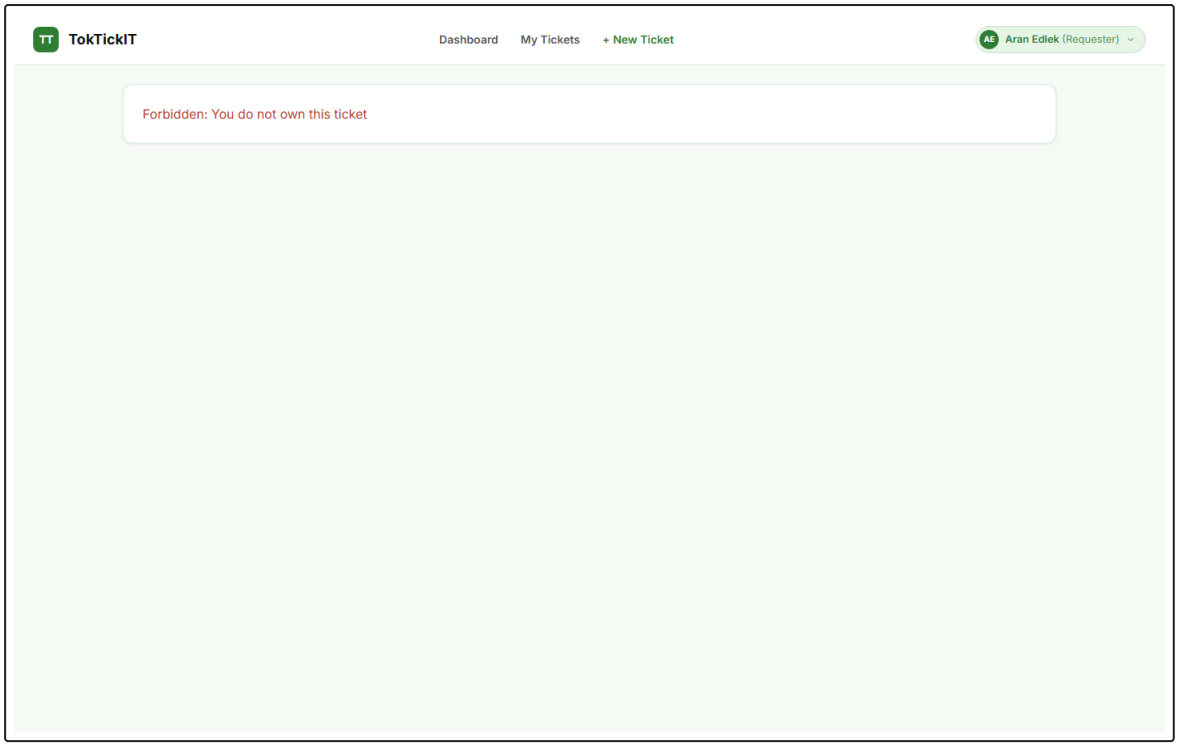


Figure 8.6: Requester Ticket Detail of TKT-2026-900002.

8.7 Regression - Attachments



Figure 8.7: Attachments - a file uploaded and listed on the Ticket.

8.8 Regression - Public Comments

IT Staff One (1/11/2026, 21:30): Please verify if your Wi-Fi reconnects after restarting the router.

Aran Edlek (Requester) (1/11/2026, 21:35): It works now, thank you!

Figure 8.8: Public Comments - the staff reply and a new comment from the Requester.

8.9 Regression - IT Staff Functions

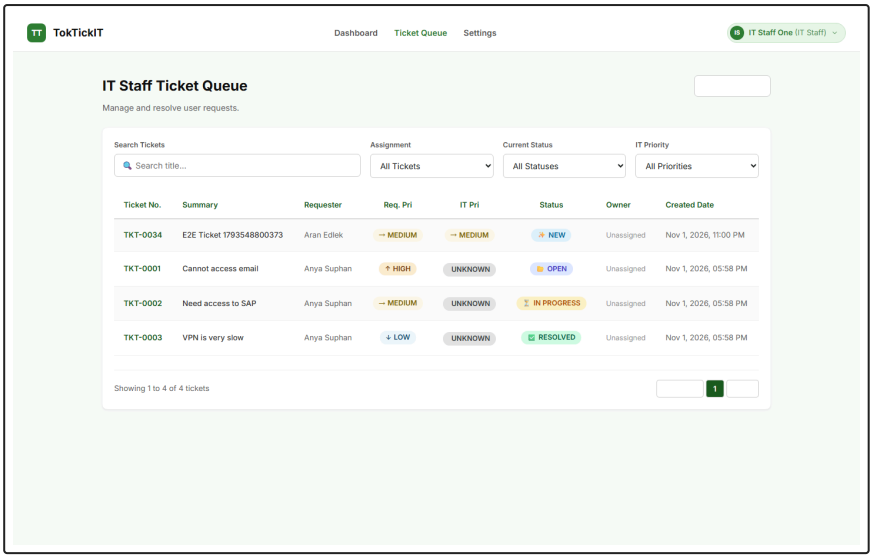


Figure 8.9: IT Staff Ticket Queue with filters, and owner and IT Priority on Staff Ticket Detail.

8.10 Regression - Internal Notes

Staff only

IT Staff One: Access point in room 8204 required firmware update. Issue resolved.

Figure 8.10: Internal Notes on TKT-2026-900002 - visible to IT Staff only.

8.11 Regression - Administrator User Management

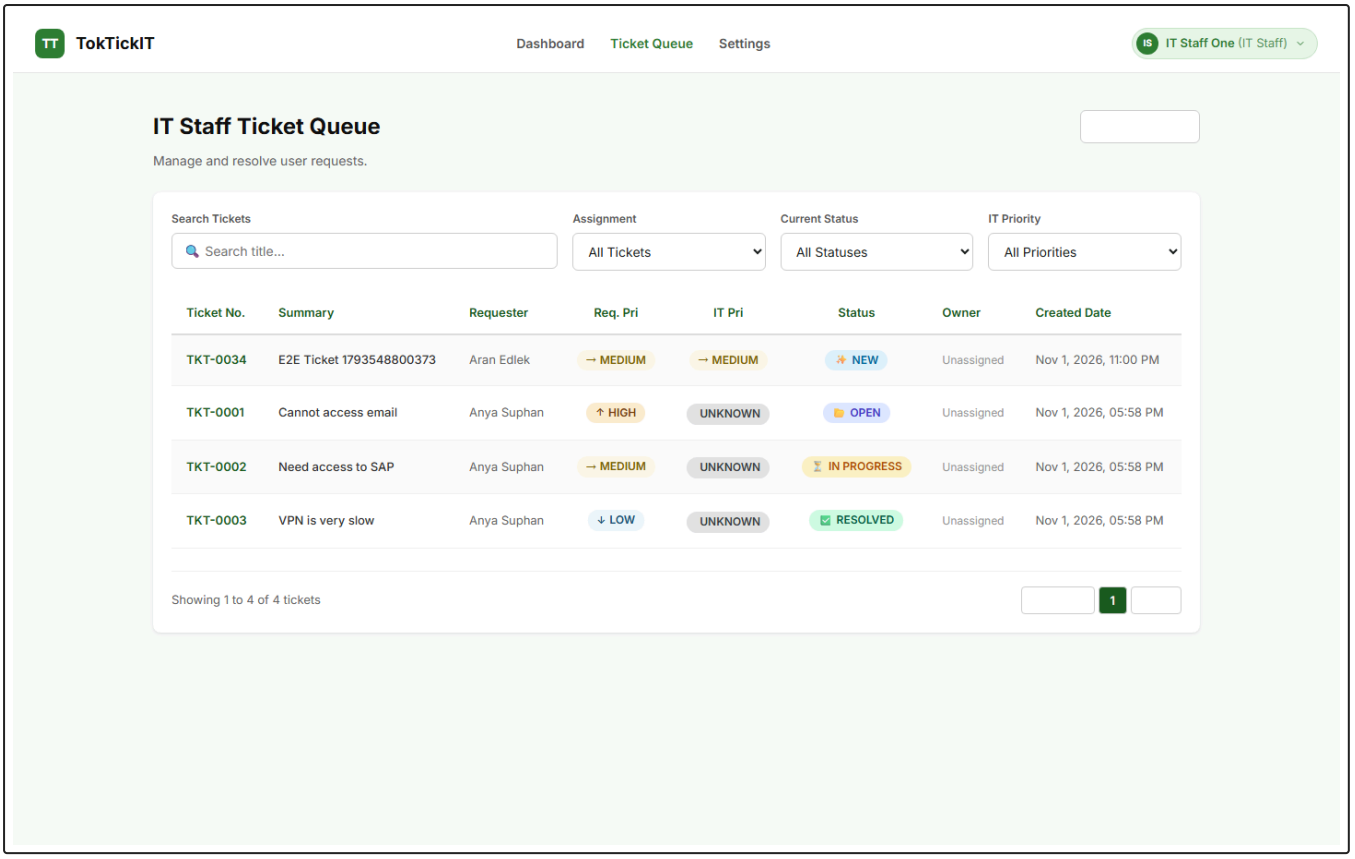


Figure 8.11: Administrator User Management - user list with roles and active status.

Answer Part 9: Zen Green UI, Responsive, Accessibility, and Final Polish

9.1 Rendered docs/lab-04/ui-spec.md

Link: https://github.com/aranedlek/toctickit/blob/main/docs/lab-04/ui-spec.md

1. Scope and References

Screen	Change
Shell and navigation	new Dashboard item, first screen after login
IT Staff Dashboard, Requester Dashboard	new
My Tickets, Ticket Queue	accept drill-down filters from the dashboard
Staff Ticket Detail	Actions Taken area, gate message, stale update message, Status Timeline
Requester Ticket Detail	read-only Actions Taken and Status Timeline

Figure 9.1: Rendered ui-spec.md (Part 1: Scope, References, Navigation)

3. IT Staff Dashboard

Card	Label	Example	Click
`unassigned`	Unassigned	"Unassigned: 3"	Ticket Queue with `ownerId=unassigned&statusGroup=active`
`myOpenTickets`	My open tickets	"My open tickets: 2"	Ticket Queue with `ownerId=:myId&statusGroup=active`
`myOpenActions`	My open actions	"My open actions: 1"	moves focus to My open actions list below

4. Requester Dashboard

Card	Label	Example	Click
`openTickets`	Open tickets	"Open tickets: 5"	My Tickets with `statusGroup=active`
`waitingForMe`	Waiting for me	"Waiting for me: 0"	My Tickets with `status=WAITING_FOR_REQUESTER`

5. Actions Taken on Ticket Detail

A new "Actions Taken (n)" card on Staff Ticket Detail after Ticket Information. It has an "+ Record Action Taken" button.

- Desktop and tablet: a table with Action Date, Description, Status, Follow-up, Assigned To, Performed By, and Edit button.
- Mobile: one card per action with the same fields and a full-width View button.
- Performed By shows "Not yet performed" until the action is completed.

Figure 9.1: Rendered ui-spec.md (Part 2: Dashboard and Actions Taken Layout Rules)

7. Screen States

Screen or area	Loading	Empty	Forbidden	Failure
IT Staff Dashboard	"Loading dashboard..."	0 cards and empty list texts	Lab 3 forbidden message	"Unable to load the dashboard." with Retry
Requester Dashboard	"Loading dashboard..."	0 cards and empty list texts	Lab 3 forbidden message	"Unable to load the dashboard." with Retry
Actions Taken	"Loading actions..."	"No actions have been recorded yet."	not used	"Unable to load the actions." with Retry

8. Responsive Rules

Same as Labs 2 and 3: Desktop 1440 × 900, Tablet 768 × 1024, Mobile 375 × 812. Tables become cards below 768 px. Long text wraps. There is no horizontal page scroll, clipping, or overlap.

9. Accessibility

- Cards and status rows are real buttons (Tab, Enter, Space) with names like "Open tickets: 3".
- Every form control has a visible label. Required fields show * and have aria-required="true".
- Statuses, badges, and messages always show text, not only colour.
- Focus is visible on every control. When the action panel closes, focus goes back to the button that opened it.

Figure 9.1: Rendered ui-spec.md (Part 3: States, Responsive, Accessibility)

11. Visual and Accessibility Checklist

Item	Screen	Viewport	Result
Design consistency with Labs 2 and 3 (colours, buttons, cards, spacing)	all Lab 4 screens	all	Pass ✅
Dashboard item shown and marked active	shell	desktop, mobile	Pass ✅
Cards show label and value; names include the value	both dashboards	all	Pass ✅
Status block and lists are readable	IT Staff Dashboard	all	Pass ✅
Actions: table on desktop and tablet, cards on mobile	Staff Ticket Detail	all	Pass ✅
Keyboard reach and visible focus	all Lab 4 screens	desktop	Pass ✅
No clipping, overlap, or horizontal overflow	all Lab 4 screens	all	Pass ✅

Figure 9.1: Rendered ui-spec.md - navigation, dashboards, Actions Taken, screen states, responsive rules, and accessibility.

9.2 IT Staff Dashboard

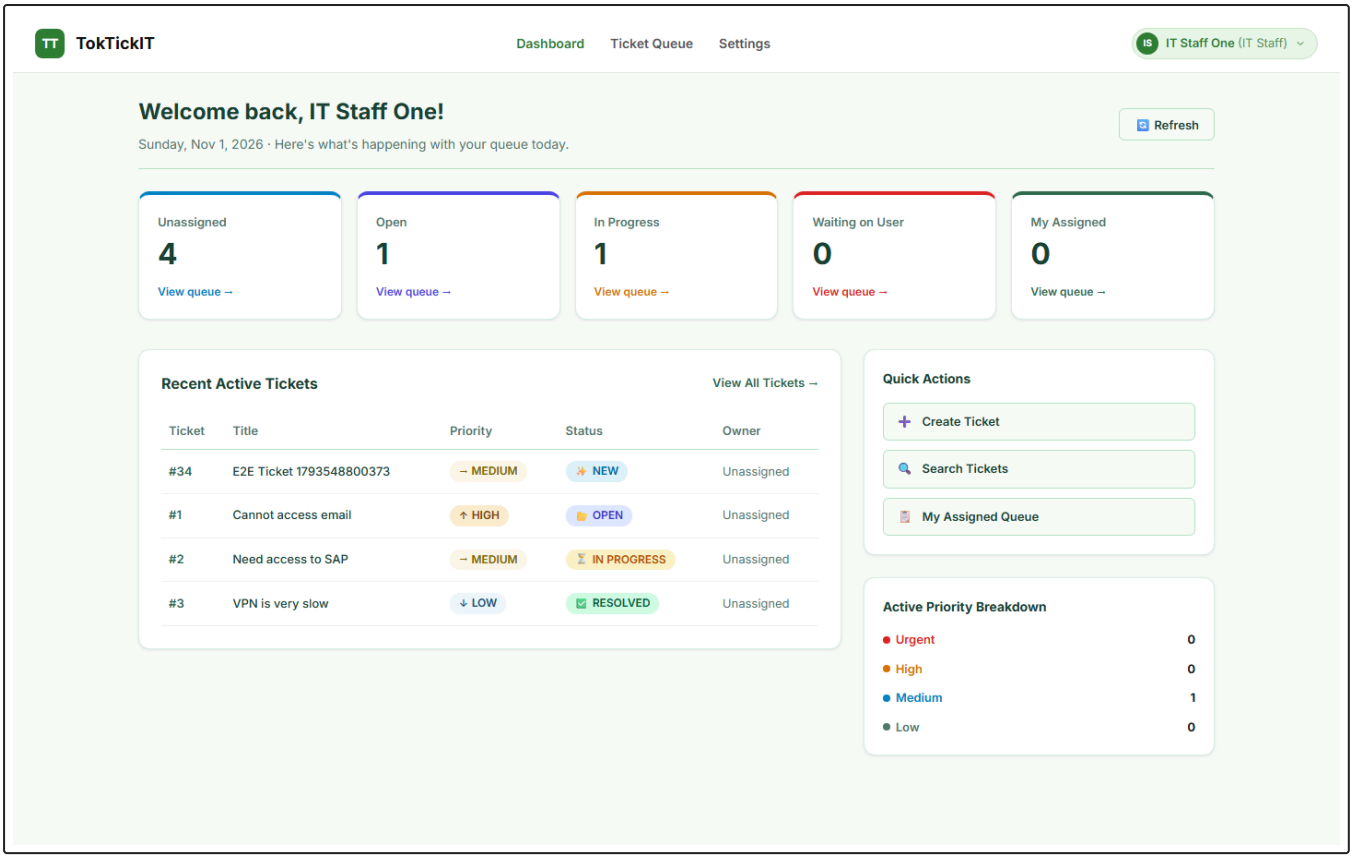


Figure 9.2a: IT Staff Dashboard on desktop (1440 × 900).

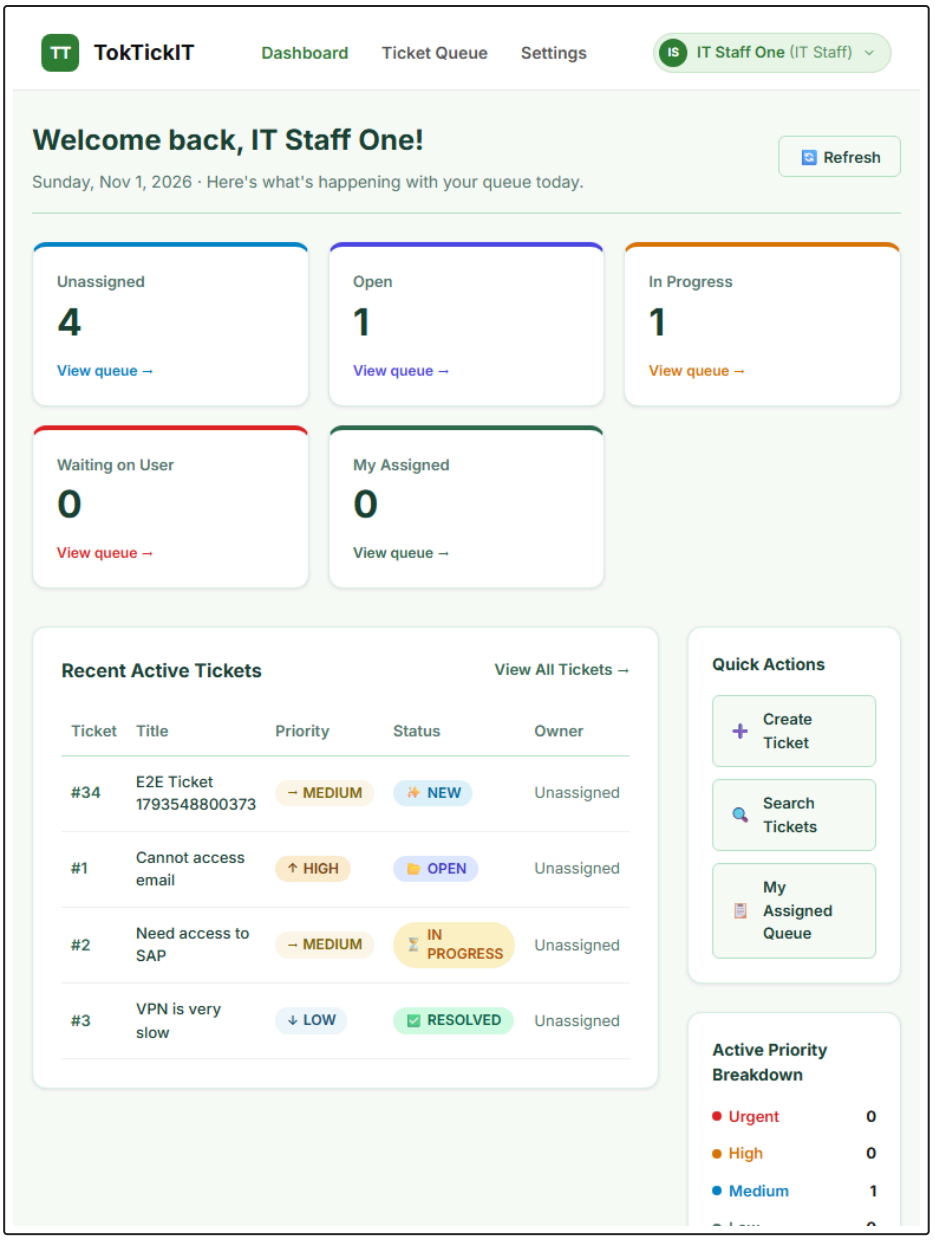


Figure 9.2b: IT Staff Dashboard on tablet (768 × 1024).

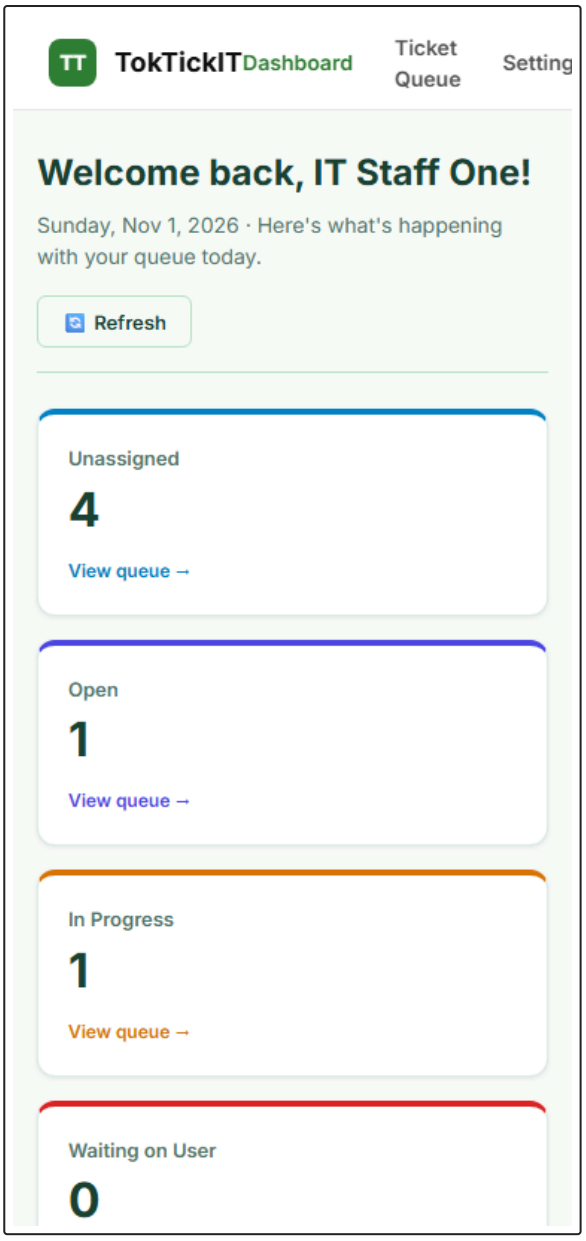


Figure 9.2c: IT Staff Dashboard on mobile (375 × 812).

9.3 Requester Dashboard

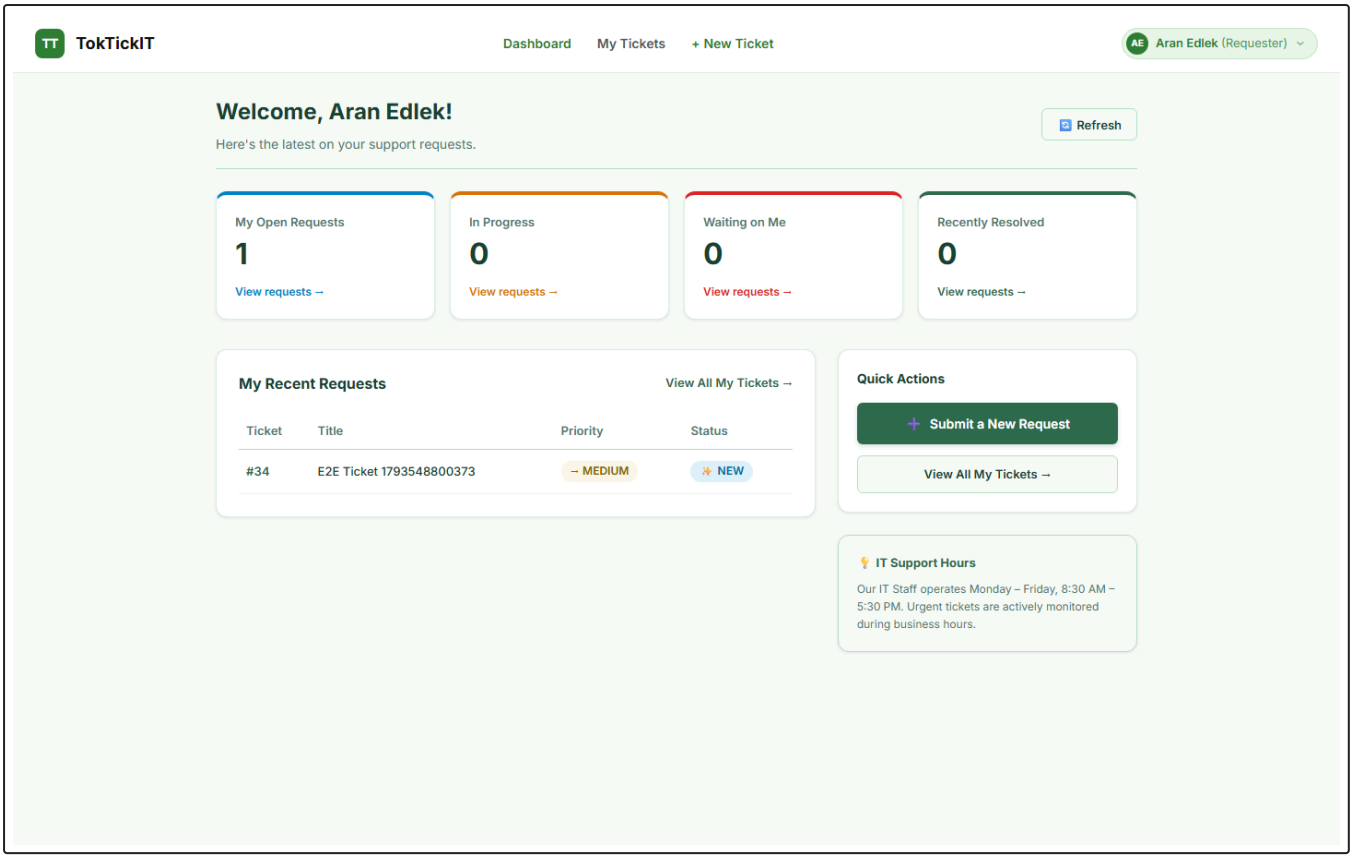


Figure 9.3a: Requester Dashboard on desktop (1440 × 900).

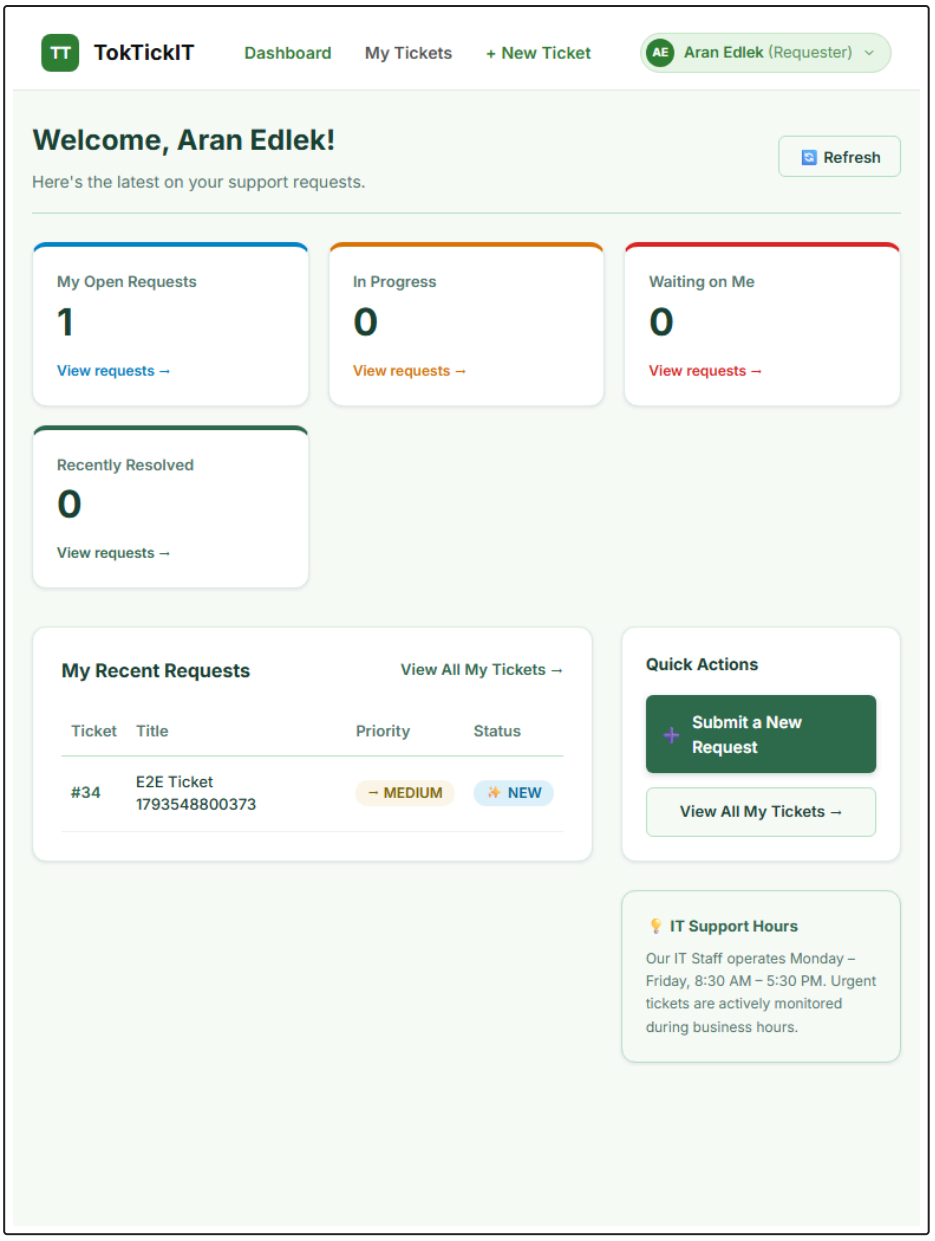


Figure 9.3b: Requester Dashboard on tablet (768 × 1024).

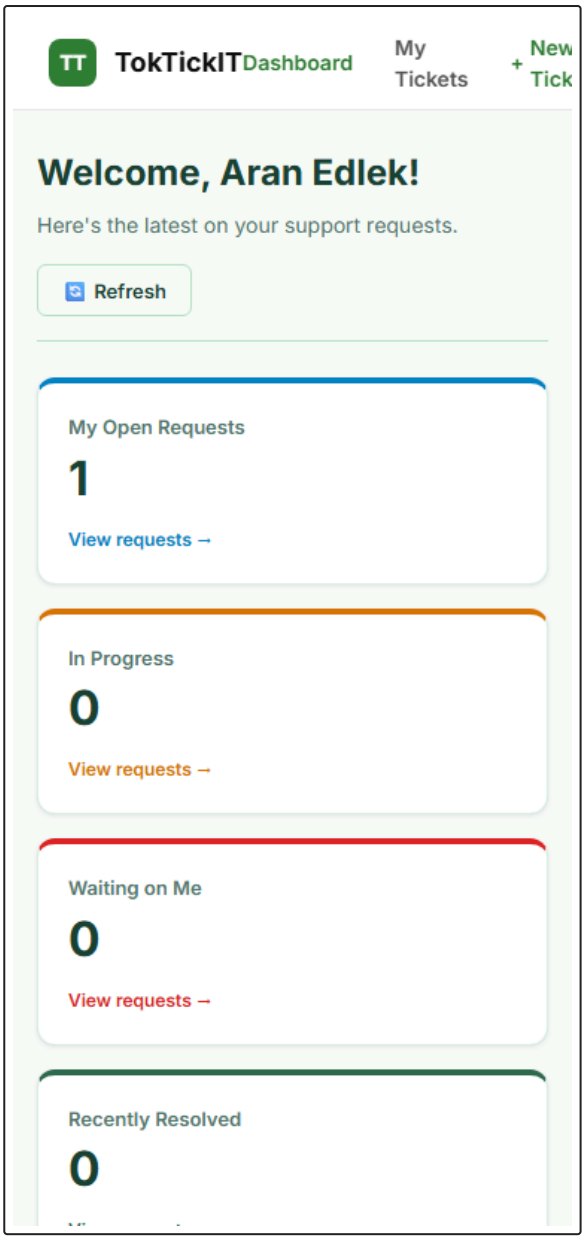


Figure 9.3c: Requester Dashboard on mobile (375 × 812).

9.4 Actions Taken on Ticket Detail

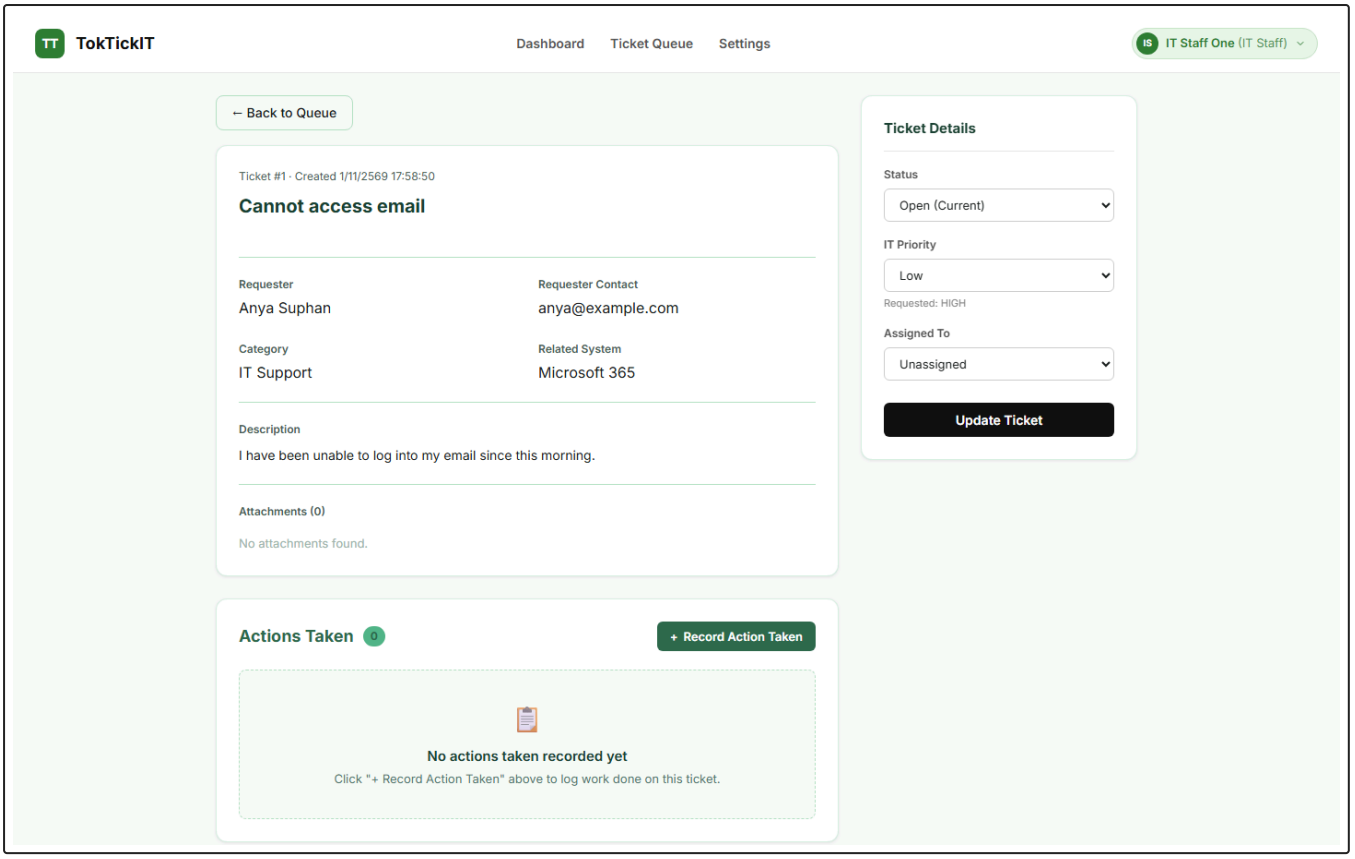


Figure 9.4a: Actions Taken on desktop

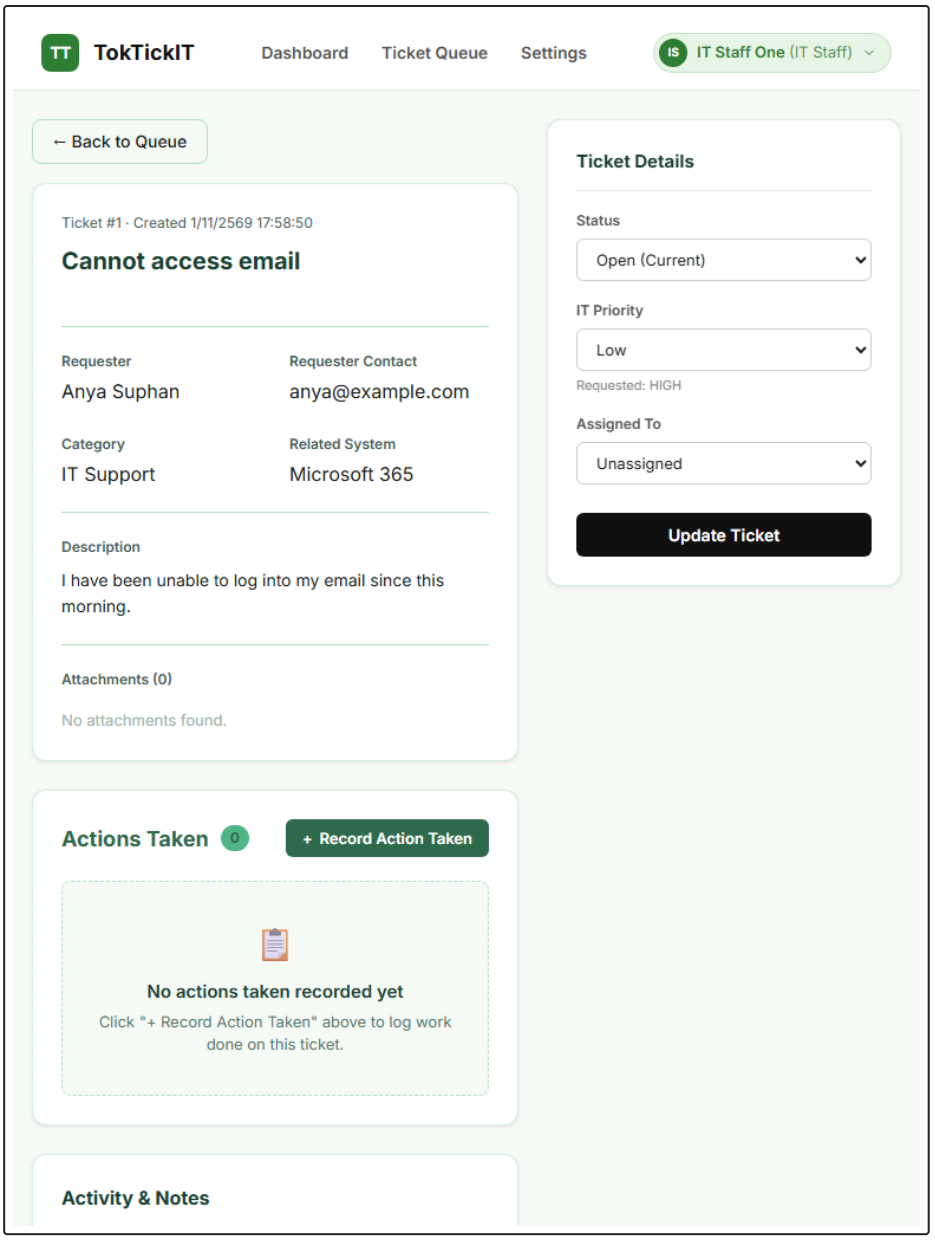


Figure 9.4b: Actions Taken on tablet

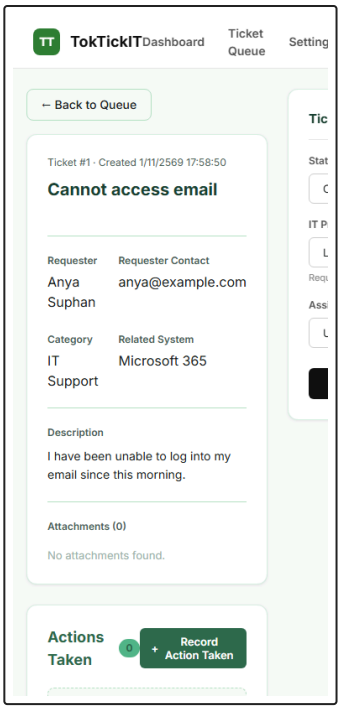


Figure 9.4c: Actions Taken on mobile

9.5 Keyboard Focus

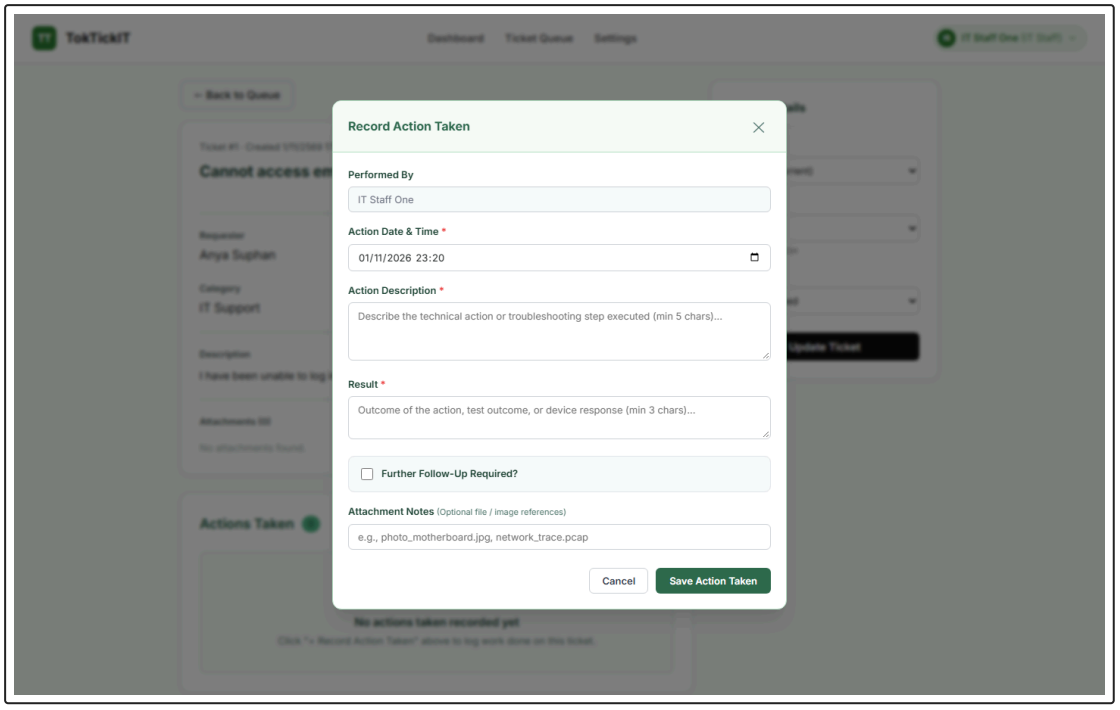


Figure 9.5: Visible keyboard focus ring on the Action Date and Time field.

9.6 Visual and Accessibility Checklist

11. Visual and Accessibility Checklist

I fill in Result after inspection using the screenshots above.

Item	Screen	Viewport	Result
Design consistency with Labs 2 and 3 (colours, buttons, cards, spacing)	all Lab 4 screens	all	Pass ✅
Dashboard item shown and marked active	shell	desktop, mobile	Pass ✅
Cards show label and value; names include the value	both dashboards	all	Pass ✅
Status block and lists are readable	IT Staff Dashboard	all	Pass ✅
Actions: table on desktop and tablet, cards on mobile	Staff Ticket Detail	all	Pass ✅
Editable and read-only fields look different	Actions Taken panel	desktop, mobile	Pass ✅
Validation under the field; other errors at the top	Actions Taken panel	desktop, mobile	Pass ✅
Gate and stale messages are clear and keep input	Staff Ticket Detail	desktop	Pass ✅
Status, priority, and private vs shared content look different	Ticket Detail	desktop, mobile	Pass ✅
Keyboard reach and visible focus	all Lab 4 screens	desktop	Pass (Fixed: Action Date and Time has focus ring) ✅
No clipping	all Lab 4 screens	all	Pass ✅
No overlap	all Lab 4 screens	all	Pass ✅
No horizontal overflow	all Lab 4 screens	all	Pass ✅

Figure 9.6: Completed checklist - design consistency, dashboards, Actions Taken, editable/read-only fields, validation placement, keyboard focus, clipping, overlap, and horizontal overflow, all Pass.